HYDRIS AI

HYDRIS FIELD GLASSES · PRODUCT REQUIREMENTS DOCUMENT

Hydris Field Glasses PRD

Product requirements for the Hydris Field Glasses platform and every product we are building

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1 Overview

1.1 Executive summary

Wastewater operators spend their shift walking the plant: reading instruments, checking pumps and blowers, sampling effluent, writing values on paper logs and responding to alarms that usually reach the control room before they reach the person standing next to the equipment. The records they produce are the evidence a plant shows its regulator, and much of that evidence is still paper.

Hydris Field Glasses puts the live state of the plant, the next task and the approved procedure in the operator's line of sight, hands free, and turns every reading, check and paper record they touch into a trustworthy digital record. It is one platform with eight products (website and demo, operator console, glasses app, Hydris Field Android, Ray-Ban Meta integration, audit scanning, voice assistant, platform API and data) that share one API, one tenant model and one permission matrix.

The platform core is **Built** and running at `https://hydrisview.vercel.app` in demo mode on synthetic plant data. The Android app and the Ray-Ban Meta paths are **Partial**: built in code, not yet validated on physical glasses. This PRD asks leadership to approve three things: the pilot scope (one or two Indian ETPs or STPs in Q1 2027), the hardening work that must land before it, and the success metrics the pilot will be judged on.

8	143	67	Q1 2027
products one API, one tenant model	**API routes** 15 modules in apps/api/src/modules	**automated tests passing** 26 domain + 41 API, local run, 22 Oct 2026	**customer pilot** target, one or two Indian plants

> **DECISION**
> Hydris is read-only towards the plant. It never writes to PLCs, SCADA or equipment, and every assistant tool is read-only (`apps/api/src/modules/voice-assistant.ts`). This keeps Hydris outside the plant's control-system safety case and is a precondition for every requirement in this document.

1.2 Problem statement

Effluent, sewage, water and common effluent treatment plants (ETPs, STPs, WTPs and CETPs) run continuously, with shift operators walking fixed rounds through intake, aeration, clarification, disinfection, sludge handling and the discharge point. Four problems recur:

1. **Paper rounds drift from reality.** Readings are written on clipboards and transcribed later. Nobody can tell which value was observed and which was copied.
2. **Alarms reach the wrong person.** The operator nearest the pump finds out last, and "acknowledged" gets confused with "fixed", so the record shows alarms cleared that were never resolved.
3. **Compliance evidence is unsearchable.** Logbooks, lab sheets and checklists are paper. Producing evidence for an inspection means digging through files.
4. **Phones are the wrong tool at the asset.** Gloves, ladders, wet hands and sampling bottles make phones awkward, and there is no industrial software for smart glasses.

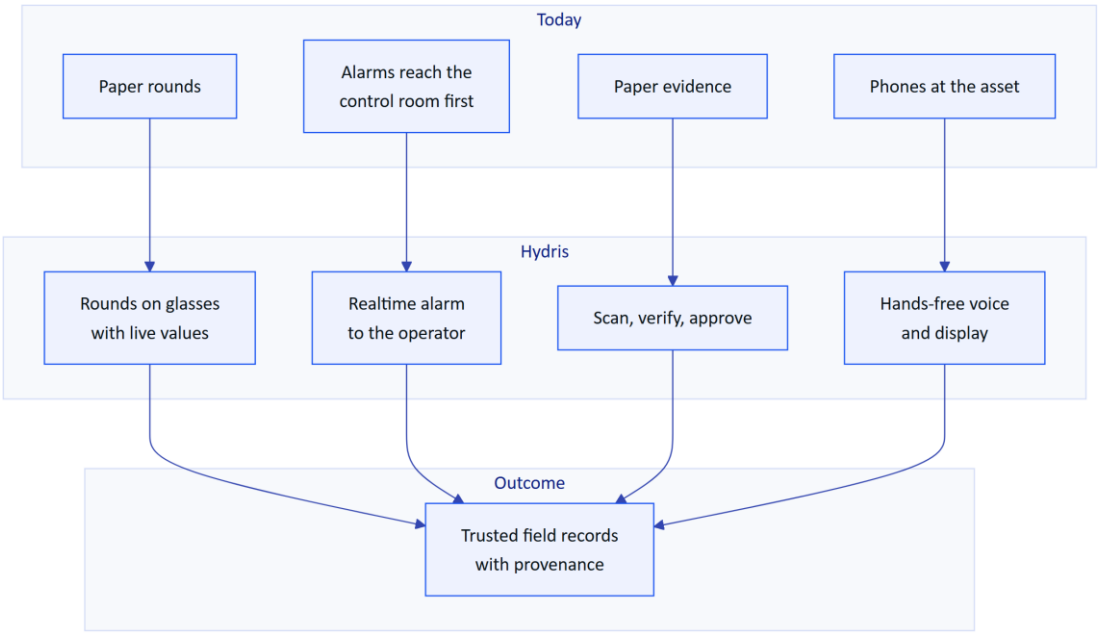


Figure 1. *From today's problems to the outcomes Hydris targets*

1.3 Opportunity and market context

Hydris is designed first for India, where discharge monitoring is a legal obligation and record quality matters:

- Discharges are regulated under the Water (Prevention and Control of Pollution) Act, 1974, through consents to establish and operate issued by State Pollution Control Boards (SPCBs) and Pollution Control Committees.
- General discharge standards are set under the Environment (Protection) Rules, 1986, with plant-specific limits in each consent.
- In 2014 the Central Pollution Control Board (CPCB) directed the 17 categories of highly polluting industries, and CETPs, to install online continuous effluent and emission monitoring systems (OCEMS) connected to CPCB and SPCB servers.

The same need (continuous monitoring, retrievable records, who did what and when) exists under NPDES permits in the United States and the Urban Waste Water Treatment Directive in the EU, which makes those later markets.

Three things make this the right time. Ray-Ban Meta camera glasses are light, affordable and acceptable on a plant floor. Meta's Wearables Device Access Toolkit (DAT 1.0.1) lets third-party Android apps use the glasses' camera, microphone, capture button and speakers. Language models can now answer spoken questions from structured plant data well enough to be useful, provided they are grounded and constrained.

Table 1. Alternatives a plant uses today and the gap Hydris fills

Alternative	Strength	Gap Hydris fills
Paper logbooks	Familiar, no devices	No provenance, no search, transcription errors
SCADA and HMI	Authoritative live data	Not at the asset; no rounds or paper digitization
Mobile forms apps	Cheap digital checklists	Not hands-free; no live context or alarm lifecycle
Remote-assist glasses apps	Expert sees what the operator sees	No plant data model, rounds or records
CMMS and EAM suites	Maintenance planning	Desktop-first, weak in the field, no glasses

> **NOTE**
> Hydris is not a certified OCEMS and does not replace OCEMS reporting to CPCB or SPCB servers. A plant's OCEMS analyzers can feed Hydris through the gateway ingestion API (FR-022) so operators see the same values the regulator sees.

1.4 Vision

Every operator in a water or wastewater plant sees what matters, does the next right thing, and leaves behind a record that stands up to an audit, without putting down their tools.

In three years we want Hydris to be the default way an Indian ETP, STP or CETP runs its rounds, responds to alarms and keeps its compliance records, and a credible choice in other markets with similar discharge-monitoring obligations.

1.5 Product principles

These principles settle design arguments. When two requirements conflict, the higher principle wins.

Table 2. Product principles and how the product applies them

ID	Principle	What it means in practice
P1	Safety first, read-only towards the plant	Never issue a control command; point to the approved procedure and the supervisor
P2	Never show a guess as a value	Missing data is "no data", never zero; stale values are marked and excluded from roll-ups
P3	Every number has provenance	A reading states its source (instrument, manual, lab, simulator) and time
P4	History is never overwritten	Corrections append; the original stays visible
P5	Humans verify, machines assist	AI and OCR propose, a person confirms, a different person approves
P6	Glance, then act	One question per glasses screen; spoken answers of one or two sentences
P7	Hands busy, still in control	Every action works with directional input or voice; the camera runs only on intent
P8	Isolation enforced by the database	Row-level security on every tenant table
P9	Synthetic data never reaches a customer	Simulator readings are refused for non-demo organizations
P10	One account everywhere	Same user, permissions and data on console, glasses and phone

2 Goals, non-goals and success metrics

2.1 Business and user goals

Table 3. Goals for the pilot and general availability

ID	Type	Goal	Evidence of success
G1	Business	Win a paid pilot at one or two Indian plants in Q1 2027	Signed pilot contract and customer sign-off at exit
G2	Business	Reach general availability in Q2 2027	Store or MDM release approved; security review passed
G3	User	Operators record rounds faster and more accurately than on paper	Time per inspection 30 % below the paper baseline (target)
G4	User	The right person acts on an alarm quickly, and closes it properly	Median time to acknowledge under 5 minutes for critical and high (target)
G5	User	Paper records become verified, searchable evidence	Median capture to approval under 24 hours (target)
G6	Trust	No answer or record presents a guess as a fact	Zero hallucinated numbers; zero cross-tenant incidents

2.2 Non-goals

- **Plant control.** No writes to PLC, SCADA or equipment, and no setpoint changes.
- **A SCADA, DCS or historian.** Hydris reads from them (connectors Planned) and never replaces them.
- **Certified regulatory reporting.** No certified OCEMS reporting to CPCB or SPCB servers.
- **A general chatbot.** The assistant answers only from the tenant's own data and approved documents, never the open internet.
- **Autonomous decisions.** AI never approves a record or generates a procedure on its own.
- **A LIMS or CMMS.** Lab results are stored as telemetry with `source = lab`; integrations come later.
- **Custom hardware or iOS before GA.** The pilot runs on Ray-Ban Meta through Android.
- **Self-serve signup for customer tenants.** Self-serve exists only in demo mode.

2.3 North-star metric

Verified field records per active operator per week: inspections submitted, alarms acknowledged within target and audit documents approved, divided by the operators who used Hydris that week. It rises only when operators use the product in the field and the records it produces are trusted.

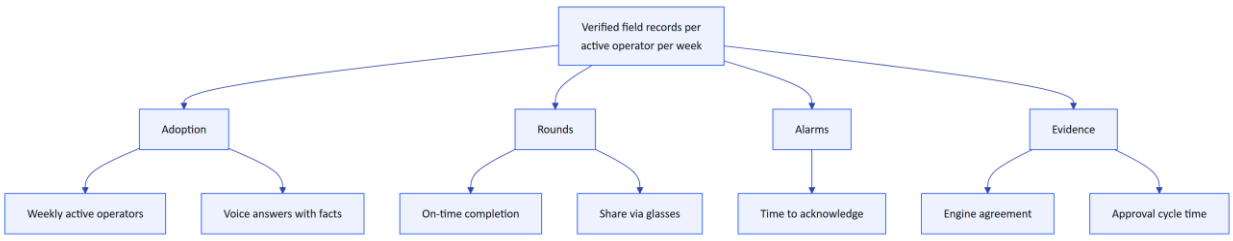


Figure 2. *KPI tree: the north star and the input metrics that drive it*

2.4 KPI table

All values below are pilot targets, not measurements. Most have a data source today in `/reports/:kind` (`apps/api/src/modules/insights.ts`); voice and agreement metrics need lightweight logging that is Planned for the pilot (NFR-034).

Table 4. KPIs with definitions, pilot targets and measurement

Metric	Definition	Target	Measurement
Weekly active operators	Operators with at least one record in a week, over licensed operators	70 % or more	Inspections and alarms reports
On-time rounds	Inspections submitted by their due time	95 % or more	`onTimeRate`, inspections report
Time per inspection	Median start to submit versus paper baseline	30 % lower	Inspections report plus baseline study
Share via glasses	Inspections submitted from glasses	50 % or more	`viaGlasses`, inspections report
Time to acknowledge	Median minutes, critical and high alarms	Under 5 minutes	`medianMinutesToAcknowledge`
Engine agreement	Average agreement score of approved documents	90 % or more	Audit document agreement score
Reviewer edits	Corrected fields over reviewed fields	Under 5 %	Audit document review data
Approval cycle	Capture to approval, median	Under 24 hours	Audit document timestamps
Voice answered with facts	Questions with a known intent	85 % or more	Intent logging (Planned)
Hallucinated numbers	Numbers in answers absent from facts	0	Automated check on logged answers
Cross-tenant incidents	Confirmed exposure across tenants	0	Security incident log
Device revocation time	Revoke to `session.ended` on the stream	30 s or less	Realtime event timestamps

3 Users and use cases

3.1 Personas

Six personas map one to one onto the roles in `packages/domain/src/roles.ts` (6 roles, 38 permissions); the Hydris platform admin is a flag on the user.

Table 5. Personas, goals, pain points and primary surface

Persona	Role	Goals	Pain points	Primary surface
Field operator (Ravi)	`field_operator`	Next task, live values at the asset, record readings, acknowledge alarms	Gloves and wet hands, copied readings, blamed for others' values	Glasses, Android app
Shift supervisor (Meena)	`supervisor`	Dispatch rounds, triage alarms, review inspections, verify scans	Cannot see late rounds or who owns an alarm	Console, Android app
Maintenance engineer (Arjun)	`maintenance_engineer`	Work orders, checklists, resolve alarms, SOPs at the asset	Manuals and history are in the office, not at the pump	Android app, glasses
Plant manager (Priya)	`factory_manager`	KPIs, alarm rules, approve documents, control glasses features	Accountable to the SPCB with paper evidence	Console
Compliance auditor (Mr Rao)	`viewer` plus `report.read`	Tamper-evident trail, approved records, exports	Cannot trace who changed what and when	Console, PDF exports
Organization admin	`org_admin`	Members, invitations, sites, glasses UI profiles	Device onboarding and revocation take too long	Console

3.2 Top use cases

Table 6. Top use cases and the requirements that serve them

ID	Use case	Served by	Status
J1	Start a round with assigned inspections in order	FR-040, FR-041, FR-059	Built on glasses; Android Planned (FR-044)
J2	See live readings and their freshness at the asset	FR-021, FR-026, FR-095	Built
J3	Acknowledge a nearby alarm and see the action and SOP	FR-031, FR-032, FR-033, FR-112	Built
J4	Capture a paper log once and get it verified and approved	FR-060 to FR-069	Built
J5	Ask by voice and hear a short, grounded answer	FR-071, FR-072, FR-074	Built (English); glasses mic Partial
J6	Control what each role sees on the glasses without an app update	FR-054	Built
J7	Pair, reassign, suspend or revoke glasses in seconds	FR-050 to FR-053, FR-057	Built; DAT registration Partial
J8	Trace every approved document and change to a person and time	FR-065, FR-066, FR-082	Built

ID	Use case	Served by	Status
J9	Try the product in minutes without talking to sales	FR-083, FR-091, FR-102	Built

3.3 Operator-day journey

The journey below is the target state for a field operator across one shift. Scores (1 to 5) are design targets, not measurements.

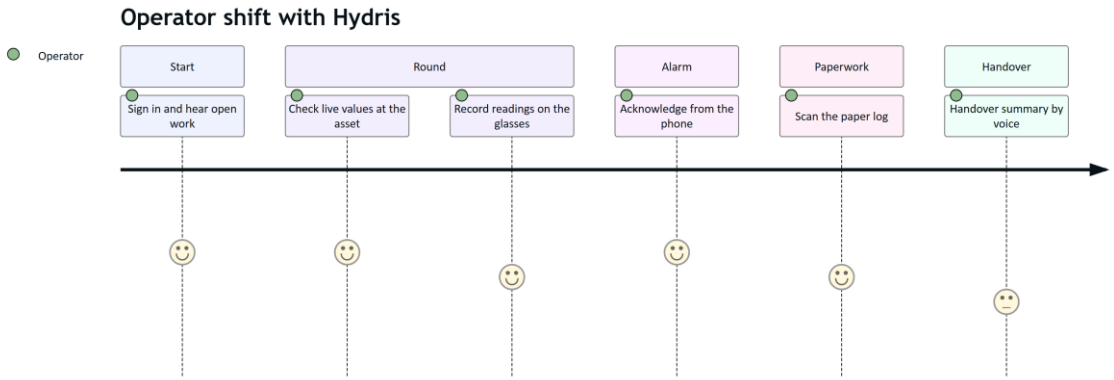


Figure 3. *Field operator journey across one shift (target state)*

The product is strongest today at alarm acknowledgement, scanning and voice lookups. The weakest step is handover: the handover summary (FR-122) is **Planned**, and inspection execution on the Android app (FR-044) is **Planned**, so rounds run in the glasses app today.

4 Solution overview

4.1 The product family

Hydris is one platform with eight products. A user signs in once and gets the same permissions and data on the console, the glasses and the phone. Every product talks to one API, and the database, not application code, enforces tenant isolation.

Table 7. The product family, sub-products, users and status

Product	Sub-products	Primary users	Status
Website and demo	Seven public pages, contact and demo request, live glasses demo, named demo accounts	Prospects	Built
Operator console	Telemetry, alarms, inspections and maintenance, plant map, devices and UI profiles, audit documents, documents and SOPs, reports, admin	Supervisors, managers, auditors, admins	Built
Glasses app	Twelve heads-up screens, phone layout, browser simulator, offline inspection queue	Field operators, engineers	Built (web); native display rendering Planned
Hydris Field Android	Sign-in, home, work, alarms, sensors, audit scan with on-device OCR, "Hey Hydris", glasses connection	Field operators, supervisors, engineers	Partial: built in code, glasses paths not validated on hardware
Ray-Ban Meta integration	Camera, capture button, microphone and speakers through DAT 1.0.1; registration as a Hydris device	Field operators	Partial: not validated on physical glasses
Audit scanning	Guided A4 capture, dual OCR, optional vision engine, field review, four-eyes approval, searchable PDF	Operators, supervisors, managers, auditors	Built
Voice assistant	Wake phrase, read-only tools, short spoken answers, rules fallback	Field operators, engineers	Built (English); other languages Planned
Platform API and data	143 routes, row-level security, realtime events, append-only audit trail, demo mode	All products	Built

4.2 System context

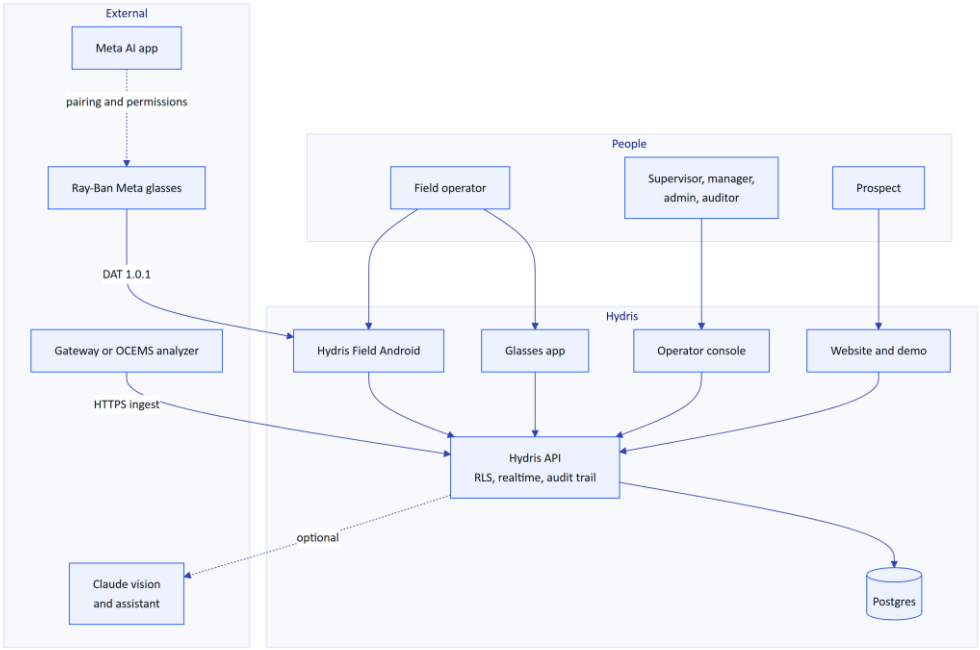


Figure 4. *System context: people, Hydris products and external systems*

Three external dependencies shape the design. Ray-Ban Meta glasses reach third-party apps only through DAT, and they are paired and permissioned through the Meta AI app, which must stay installed on every operator phone. Plant data arrives over HTTPS from an edge gateway or OCEMS analyzer (FR-022); MQTT and OPC UA adapters are **Planned**. The Claude model is optional: without it, a rules engine answers questions and OCR alone extracts documents (NFR-037).

4.3 The core loops

Four loops explain the whole product. Each crosses several products, which is why they share one platform.

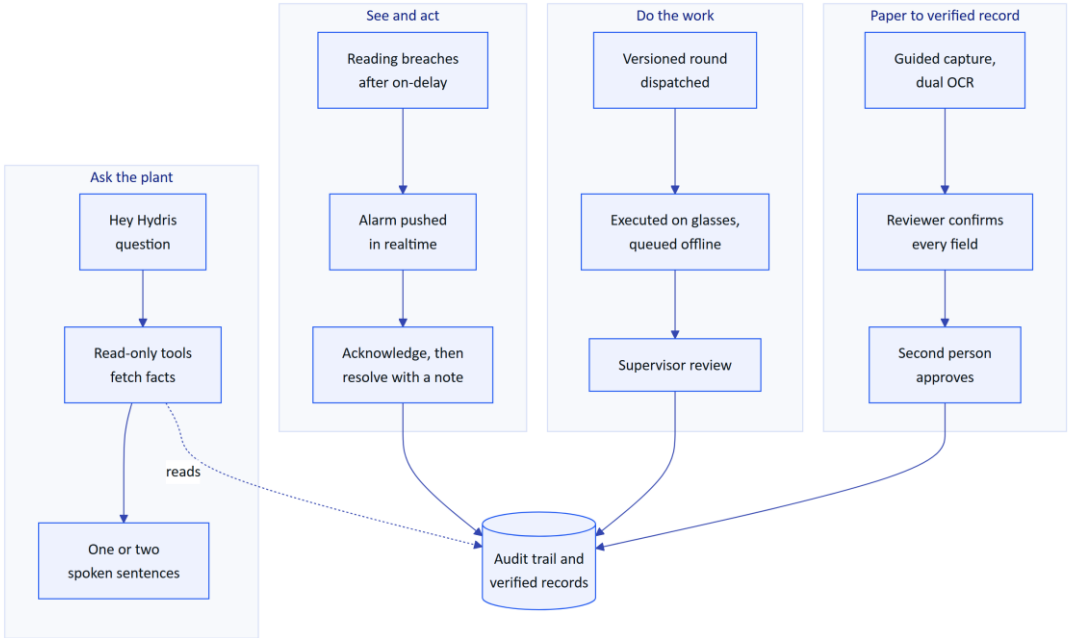


Figure 5. *The four core loops and the record each one produces*

Table 8. The core loops, what makes them trustworthy, and status

Loop	What makes it trustworthy	Status
See and act	Deadband and on-delay stop flapping; an alarm cannot be resolved without a note, so acknowledged is never mistaken for fixed	Built; Android Resolve fix in Hardening (FR-094)
Do the work	Templates are versioned; responses queue on the device and replay exactly once	Built on glasses; Android inspections Planned (FR-044)
Paper to verified record	The original extracted value is kept beside every correction; approval must come from a different person	Built
Ask the plant	Answers use only tool results from the same conversation and state the age of each reading; unknown questions get "can't answer"	Built (English)

4.4 Release scope by product

The MVP is complete in code. The pilot adds the hardening and hardware validation needed to run at a real plant; GA and later phases broaden reach.

Table 9. Release scope by product, pilot versus later

Product	Pilot (Q1 2027)	Later (GA Q2 2027 and Advanced)
Website and demo	As built	Content updates only
Operator console	Compliance summary (FR-120), alarm escalation (FR-037), email transport (FR-086)	SSO (FR-011), recurring schedules (FR-047)
Glasses app	As built, on the web runtime and phone layout	Native glasses display rendering
Hydris Field Android	Resolve fix (FR-094), inspections (FR-044), FR-099 fixes	FCM push (FR-034), offline outbox (FR-098), Play or MDM distribution
Ray-Ban Meta integration	Hardware validation of camera, capture button, speech, registration (FR-055 to FR-057, FR-074) in Developer Mode	Store or Developer Center distribution
Audit scanning	Page images moved to object storage	E-signatures (FR-068), LIMS
Voice assistant	English, rules or model engine per tenant consent	Hindi, Tamil, Marathi, Gujarati (FR-073)
Platform API and data	CI (NFR-010), job cadence (NFR-008), observability (NFR-034)	India-region production (NFR-017), pen test (NFR-018), MQTT and OPC UA (FR-027)

DECISION

The pilot runs on the native Android app, not the WebView bridge. The native app keeps device tokens in the Android Keystore, uses ML Kit for on-device OCR and talks to DAT directly. Every glasses feature also works on the phone, so the pilot does not depend on Meta's distribution timeline.

5 Operator console

The operator console is the web application at `/console` where supervisors, managers, auditors and administrators run the plant record. Field operators produce data with the glasses and the Android app; the console is where that data is dispatched, triaged, reviewed, approved and reported. It is one Next.js surface (`apps/web/src/console`) over the same API that serves every other client, so a value acknowledged on the glasses is the same record a manager sees on the overview a second later.

Table 10. Console sub-products

Sub-product	Route	Job it does	Status
Overview dashboard	`/console`	Is anything wrong right now, and where	Built
Plant model and map	`/console/map`, sites, assets	Describe the plant once and place it on a map	Built
Telemetry	Sensor monitoring	Honest live values with source and freshness	Built (HTTPS)
Alarms	Alarms and events	Turn readings into attributed work	Built, escalation Planned
Inspections and maintenance	Inspections, maintenance	Design, dispatch and review rounds and tasks	Built, schedules Planned
Documents and SOPs	`/console/documents`	Approved procedures at the asset	Built
Glasses fleet and UI profiles	Devices, `/console/glasses-profiles`	Pair, revoke and configure glasses	Built
Audit review and approval	`/console/audit-docs`	Turn scanned paper into approved evidence	Built, e-signature Planned
Reports	`/console/reports`	Numbers managers and auditors can defend	Built, compliance Planned
Administration and roles	Members, settings, audit log	Who can do what, with no escalation	Built, email and SSO Planned

5.1 Overview

Water and wastewater plants still run on paper log sheets, phone calls and spreadsheets. The supervisor cannot see which rounds are late, the manager cannot prove to a regulator who saw an alarm and when, and the auditor receives evidence that cannot be traced to a person. The console replaces that patchwork with one tenant-isolated, site-scoped workspace where every state change is attributed, timestamped and written to an append-only audit trail. Navigation (`Shell.tsx`) hides any destination the current role lacks, and the API enforces the same check from one fixed role matrix (`packages/domain/src/roles.ts`, six roles, 38 permissions).

5.2 Users and user stories

Table 11. Console user stories

ID	Story	Priority
US-C1	As a shift supervisor I want open alarms ranked by severity so that I triage the worst first	P0
US-C2	As a shift supervisor I want to dispatch a route to an operator and review the result so that rounds are traceable	P0
US-C3	As a shift supervisor I want to approve glasses pairing from any phone so that a new pair is live in minutes	P0

ID	Story	Priority
US-C4	As a plant manager I want threshold rules with deadband and on-delay so that operators are not flooded by noise	P0
US-C5	As a plant manager I want to approve verified audit documents so that paper logs become evidence	P0
US-C6	As a plant manager I want alarm and inspection reports in CSV so that I can defend KPIs	P1
US-C7	As a compliance auditor I want an append-only trail of who did what so that evidence is attributable	P0
US-C8	As an organization admin I want to decide what each role sees on the glasses so that screens fit the job	P1
US-C9	As an organization admin I want to invite members with a role no higher than mine so that privileges never escalate	P0

5.3 Scope

Table 12. Console scope

In scope	Out of scope
Overview, plant model and map, telemetry with freshness and corrections	User-configurable widgets, cross-site benchmarking
Alarm rules and five-state lifecycle	Any write path to plant equipment (Hydris is read-only towards the plant)
Inspection templates, routes, dispatch, review; maintenance tasks	Digital twins, P&ID import
Document library with approval; grounded SOP retrieval	Bulk import of the asset register (gap)
Glasses fleet, pairing approval, UI profiles	Fleet health signals such as battery (next step)
Audit document review and four-eyes approval	Capture and OCR (glasses and Android sections)
Reports in JSON and CSV; admin, roles, audit log	Pricing, billing

5.4 Overview dashboard

The overview is the first screen after sign-in: KPI tiles, open alarms by severity, asset health, live measurements with freshness and recent activity. It is served by `GET /overview`, the same endpoint that feeds the Android home screen, so the console and the phone never disagree. Stale values are excluded from health roll-ups (`packages/domain/src/freshness.ts`) and synthetic values carry a visible label. A command palette (`CommandPalette.tsx`) gives keyboard access to every destination and global search. Next step: a per-user default site and a compliance tile once FR-120 lands.

5.5 Plant model and map

Everything hangs off the plant model: organization, site, facility, process area, asset and sensor. The site is the permission boundary, so a site-scoped supervisor sees nothing from other sites. Archiving an asset removes it from glasses screens and assistant answers while keeping its history. The plant map (`/console/map`) places assets on the site layout and overlays route stops and open alarms; the hierarchy navigator and asset schematic give the same model as a tree and per asset. Gateways authenticate with `hgw_` API keys. Gaps: no bulk import of the asset register, and the map is drawn per site rather than imported from drawings.

5.6 Telemetry

The core promise is honesty: a missing reading is shown as "No data", never as zero, and every value carries its source (gateway, manual, lab or synthetic) and time. Gateways post up to 500 readings per call; readings are keyed on sensor and timestamp so a replayed batch never duplicates. Freshness is live up to twice the expected interval, delayed up to five times, then stale. A wrong value is never edited: a supervisor adds a correction or invalidates it, and the original stays visible with author and reason. MQTT, OPC UA and historian connectors are Planned.

5.7 Alarms

Alarms turn readings into attributed work. Acknowledging records that a person has seen the alarm; it never changes the process. Rules support HH, H, L, LL and stale conditions with deadband (no flapping around a limit) and on-delay (short spikes ignored). There is no path from active straight to resolved, and the API rejects a resolve without a note. Every transition writes a history row and an audit event, and pushes in realtime to the console, glasses and Android.

> **RISK**
> Stale rules run every 30 s on a long-running server but only once a day on the current Vercel cron (FR-035, Partial), and unacknowledged alarms do not escalate (FR-037, Planned). Both block an unsupervised pilot.

5.8 Inspections and maintenance

A template is built in the console (`TemplateBuilder.tsx`) from five step kinds: check, reading with expected range, choice with abnormal options, text and photo. Editing creates a new version, so dispatched inspections keep the version they started with. Routes order templates along a walk and are dispatched to an operator. The supervisor reviews a submitted inspection with abnormal steps highlighted and approves it or returns it with a reason; offline responses from the glasses replay idempotently, so the reviewer never sees duplicates. Maintenance tasks are corrective, preventive or follow-ups from an alarm or inspection, with priority, checklist and activity log, and move through open, assigned, in progress, on hold, completed or cancelled, with reopen. Every round is dispatched by hand today; recurring schedules are Planned.

5.9 Documents and SOPs

The library holds SOPs, manuals, datasheets, safety notices and troubleshooting guides, each with versions and a scope of site, asset or asset type. The lifecycle is draft, approved, archived, and only approved documents reach the glasses, the assistant and grounded retrieval. Alarm rules link a procedure shown with the alarm. The library page exposes grounded SOP retrieval (`/assist`), so a supervisor can check what an operator will be told before it happens. Uploading existing PDF manuals with text extraction is Planned.

5.10 Glasses fleet and UI profiles

Glasses are shared, often unattended devices, so the console makes them a managed fleet. A supervisor approves pairing to a site and an operator; the glasses never hold a user password, only a device credential the console can end. Devices move through pending, paired, assigned, connected, disconnected (not seen for 90 s), suspended and revoked. Access tokens last 15 minutes and refresh tokens rotate with reuse detection. A device can never exceed the device permission ceiling, whatever the operator's role.

A UI profile controls what the glasses show. Profiles are drafted, published and rolled back; assignment is by device, user, role or organization, and the most specific published assignment wins. Publishing pushes the change live without a restart.

5.11 Audit review and approval

Scanned paper logs become evidence only after one person has checked them and a second has approved them. The reviewer sees each extracted value beside its crop of the page, with the engines that read it and their agreement. Values where at least two engines agree with confidence 0.8 or more can be bulk-confirmed; everything else is confirmed, corrected or flagged one by one, and the original reading is always kept. Verify is blocked while any item is pending or flagged. Approval needs `auditdoc.approve` and must come from someone other than the verifier (demo tenants allow recorded self-approval). The approved PDF carries the verification trail, the searchable original and a reconstructed page.

5.12 Reports

Reports (`insights.ts`) cover inspections, alarms, maintenance, devices and operators, in JSON for the page and CSV for export. The alarm report includes median minutes to acknowledge; the inspection report includes on-time rate and the share submitted via glasses. These fields feed most pilot KPIs. A compliance summary per discharge point, the report regulators actually ask for, is Planned.

5.13 Administration and roles

The rule is no escalation: nobody can grant a role above their own (`assignableRoles`). Invitations carry a 7-day token; members can belong to several organizations and switch; sessions can be listed and revoked; TOTP MFA is available. The audit log page filters the append-only trail by action and entity, and no API path updates or deletes an audit event. Platform admins create and suspend customer organizations at `/console/platform`. Invitation emails are queued in `mail_outbox` but not yet sent.

Table 13. Role access summary (roles.ts)

Capability	Admin	Manager	Supervisor	Engineer	Operator	Viewer
Acknowledge alarms	Yes	Yes	Yes	Yes	Yes	No
Resolve / close alarms	Yes / Yes	Yes / Yes	Yes / Yes	Yes / No	No	No
Manage alarm rules, plant model	Yes	Yes	No	No	No	No
Review inspections, manage maintenance	Yes	Yes	Yes	No	No	No
Pair glasses / manage profiles	Yes / Yes	Yes / Yes	Yes / No	Yes / No	Yes / No	No
Verify / approve audit documents	Yes / Yes	Yes / Yes	Yes / No	No	No	No
Audit log	Yes	Yes	No	No	No	No

> **DECISION**
> The auditor persona maps to `viewer`, which holds `report.read` but not `audit.read`. Open decision: grant `audit.read` to viewer or add a dedicated auditor role.

5.14 Functional requirements

Priorities map the source PRD as Must to P0, Should to P1 and Could to P2.

Table 14. Console functional requirements by sub-product

ID	Requirement	Pri	Status
FR-089	Overview: KPIs, open alarms, health, live measurements	P0	Built
FR-085	In-app notifications in realtime, mark as read	P0	Built
FR-088	Global search across assets, sensors, documents and work	P1	Built
FR-020	Plant hierarchy with map positions, archive and restore	P0	Built
FR-021	Sensors for 16 measurands with unit normalization	P0	Built
FR-007	Site-scoped permissions	P0	Built
FR-022	Gateway ingestion, 500 per call, idempotent	P0	Built
FR-023	Manual and lab entry with provenance	P0	Built
FR-024	Append-only corrections (correct or invalidate)	P0	Built
FR-025	History, raw readings, CSV export	P1	Built

ID	Requirement	Pri	Status
FR-026	Freshness: live, delayed, stale, no data	P0	Built
FR-029	Synthetic data labelled, refused for customers	P0	Built
FR-027	MQTT and OPC UA adapters	P1	Planned
FR-028	Read-only SCADA and historian connectors	P2	Planned
FR-030	Threshold rules with deadband and on-delay	P0	Built
FR-031	Five-state lifecycle, no resolve from active, note required	P0	Built
FR-032	Assignment, notes, history, recommended action	P0	Built
FR-033	Realtime push to console, glasses, Android	P0	Built
FR-035	Stale-data alarms on a schedule	P0	Partial
FR-037	Escalation when unacknowledged	P1	Planned
FR-038	Shelving with reason and expiry	P2	Planned
FR-040	Versioned templates, five step kinds, routes, dispatch	P0	Built
FR-042	Submit, review, approve or return	P0	Built
FR-043	Maintenance tasks, checklists, follow-ups	P0	Built
FR-047	Recurring schedules	P1	Planned
FR-110	Document library: draft, approved, archived; versions; scope	P0	Built
FR-111	Only approved documents reach glasses and assistant	P0	Built
FR-112	Alarm rule links a procedure shown with the alarm	P0	Built
FR-070	Grounded SOP retrieval with safety notice	P0	Built
FR-114	Upload PDF manuals with text extraction	P2	Planned
FR-050	Pairing code, QR, 5 min TTL, 3 s polling, approval	P0	Built
FR-051	Device tokens, rotation, reuse detection	P0	Built
FR-052	Suspend, reactivate, revoke, reassign	P0	Built
FR-053	Device permission ceiling	P0	Built
FR-054	UI profiles: draft, publish, rollback, precedence, live push	P1	Built
FR-064	Confirm, correct, flag, reset; verify gate	P0	Built
FR-069	Bulk-confirm agreed multi-engine values	P1	Built
FR-065	Independent approval (four eyes)	P0	Built
FR-066	PDF with trail, searchable original, reconstruction	P0	Built
FR-068	Electronic signature with re-authentication	P1	Planned
FR-081	Reports for five areas in JSON and CSV	P0	Built
FR-120	Compliance summary per discharge point	P1	Planned
FR-121	Scheduled report delivery by email	P2	Planned
FR-004	Invitations, 7-day token, no escalation	P0	Built
FR-012	Fixed role matrix	P0	Built
FR-082	Append-only audit trail	P0	Built

ID	Requirement	Pri	Status
FR-087	Audit log page with filters	P0	Built
FR-010	Platform organization create and suspend	P0	Built
FR-005	Password reset email	P0	Partial
FR-086	Email delivery (SMTP, SES or Resend)	P0	Planned
FR-011	SSO with SAML 2.0 and OIDC	P1	Planned

5.15 Key flows

A new reading is normalized, stored idempotently and evaluated against rules before events reach every client.

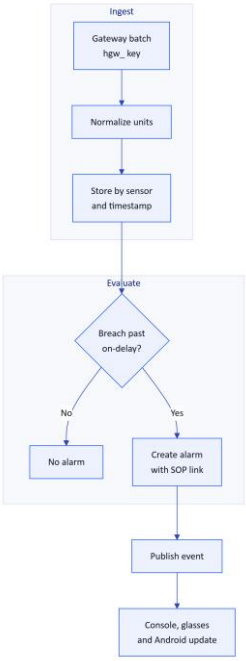


Figure 6. *Telemetry to alarm (telemetry.ts, alarms.ts, realtime.ts)*

An alarm must be seen before it can be resolved, and resolution needs a note.

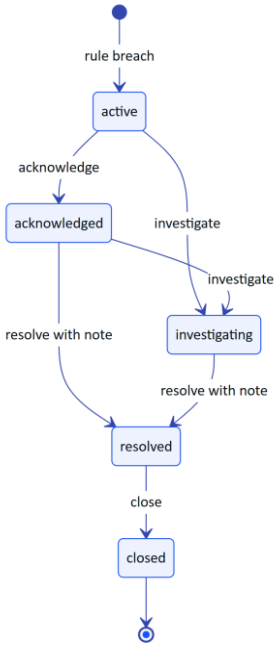


Figure 7. *Alarm lifecycle (packages/domain/src/alarms.ts)*

An inspection is dispatched, executed in the field and approved or returned by a supervisor.

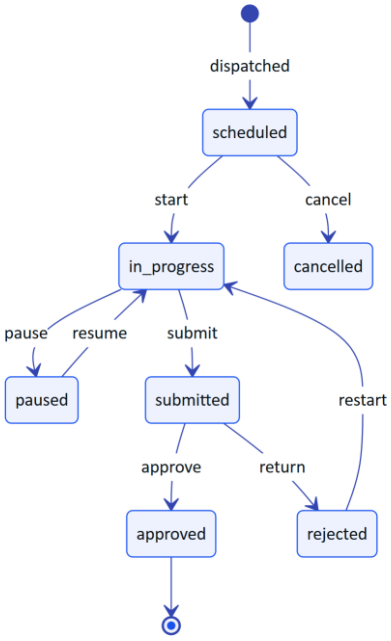


Figure 8. *Inspection lifecycle (workflows.ts)*

The glasses never type a password; a signed-in supervisor approves the code and the glasses claim their own credential.

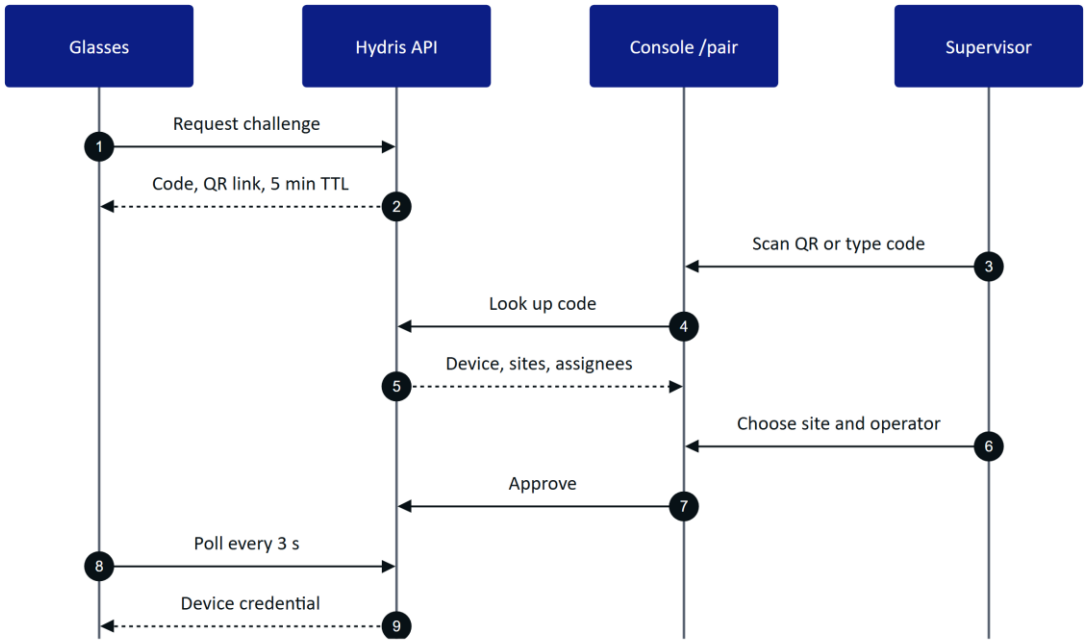


Figure 9. *Console pairing approval (devices.ts)*

5.16 Acceptance criteria

- Given an active alarm, when a user tries to resolve it directly, then the API rejects the transition; and given an acknowledged alarm, when resolve is sent without a note, then it is rejected (FR-031).
- Given a gateway batch already stored, when the same batch is replayed, then no duplicate readings are created (FR-022).
- Given a sensor with no reading, when the overview loads, then it shows "No data" and never zero (FR-026).
- Given a submitted inspection, when the supervisor returns it with a reason, then the operator can restart it and history shows both submissions (FR-042).
- Given a verified audit document, when the verifier tries to approve it in a customer tenant, then approval is refused (FR-065).
- Given a revoked device, when it makes its next request, then it is refused and its event stream closes within 30 s (FR-052).

5.17 Success metrics

All values are pilot targets, not measured results.

Table 15. Console metrics (pilot targets)

Metric	Source	Target
Median time to acknowledge, critical and high	Alarm report	Under 5 minutes
On-time round completion	Inspection report `onTimeRate`	95 % or more
Share of rounds submitted via glasses	Inspection report `viaGlasses`	50 % or more
Audit document capture to approval	Audit documents	Under 24 hours
Reviewer edits per reviewed field	Audit documents	Under 5 %
Device revocation to stream end	Event stream	30 s or less
Realtime alarm delivery	NFR-002	2 s (not yet measured)
Cross-tenant incidents	Security review	0

5.18 Dependencies and open issues

- Depends on the platform API, Postgres row-level security (FR-008) and the realtime event stream (FR-080).
- Blocks an unsupervised pilot: email delivery (FR-086), escalation (FR-037) and a reliable stale schedule (FR-035).
- Open: auditor role decision; compliance summary (FR-120); SSO (FR-011); electronic signature (FR-068); NFR-001 and NFR-002 not yet load-tested.

6 Website and demo

6.1 Overview

Industrial buyers rarely touch a product before a sales cycle starts, and smart-glasses software is hard to picture without hardware. The website explains Hydris and lets a prospect try it in minutes: drive the real glasses viewport in the browser, then sign in to the console or the Android app with just a name. Everything runs on synthetic data labelled as such. The goal is a self-serve path from curiosity to a working demo account with no sales contact.

6.2 Users and user stories

Table 16. Website and demo user stories

ID	Story	Priority
US-W1	As a prospect I want to drive the glasses UI in my browser so that I understand the product without hardware	P0
US-W2	As a prospect I want to sign in with only a name so that I can try the console immediately	P0
US-W3	As a salesperson I want my named account to survive between sessions so that demos are repeatable	P0
US-W4	As a trainer I want deterministic scenarios so that every class sees the same events	P1
US-W5	As a prospect I want to request a guided demo so that I can talk to the team	P1

6.3 Scope

Table 17. Website and demo scope

In scope	Out of scope
Seven public pages plus privacy, phone and desktop, light and dark	Self-serve sign-up for customer tenants
Contact and demo-request form	Pricing pages, payments
Client-side glasses demo with ten scenarios; browser simulator and pairing	Real plant data of any kind
One-click named demo accounts on console and Android	Guest data beyond the shared store's 30-day TTL

6.4 Functional requirements

Table 18. Website and demo requirements

ID	Requirement	Pri	Status
FR-100	Marketing website, seven pages and privacy page	P0	Built
FR-101	Contact form, 5 per 10 minutes per client	P0	Built
FR-102	Client-side glasses demo plus browser simulator	P0	Built
FR-103	Named guests with personal copies of open work	P0	Built
FR-083	Zero-configuration demo tenant, cross-instance pairing	P0	Built
FR-084	Ten deterministic scenarios, refused outside demo tenants	P0	Built
FR-091	Demo sign-in with a name in the Android app	P0	Built
FR-029	Synthetic data labelled, refused for customer tenants	P0	Built

6.5 Key flows

A guest id is derived from the normalized name, so the same name always returns to the same account, rebuilt on any serverless instance from the shared store.

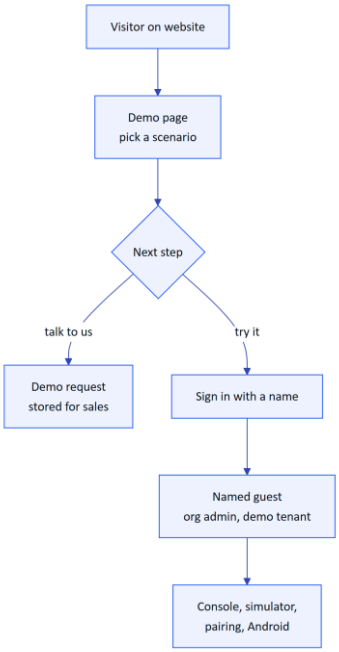


Figure 10. *From visitor to named demo account*

6.6 Acceptance criteria

- Given a deployment with no environment variables, when a visitor opens `/demo` and signs in, then the demo works end to end (FR-083).
- Given a guest named "Asha", when she returns later with the same name, then she reaches the same account and work (FR-103).
- Given two guests acting at once, when one completes an inspection, then the other's copies are unchanged (FR-103).
- Given a customer tenant, when a simulation scenario is started, then it is refused (FR-084).

> **WARNING**
> Demo mode relaxes security on purpose (12-hour access tokens, public signing secrets, audit self-approval). Production sets `DEMO_MODE=false` and `SIMULATOR_ENABLED=false`.

6.7 Success metrics

Table 19. Website and demo metrics (targets)

Metric	Target
Landing to signed-in demo account	Minutes, no sales contact
Demo on zero-configuration deployment	Works with no environment variables
Guest isolation failures	Zero
Demo requests converted to guided demos	Target to be set at pilot

6.8 Dependencies and open issues

- Inquiries are stored but no notification email is sent until email delivery (FR-086) lands.
- The Vercel Runtime Cache store is per region and best effort; Postgres is preferred where available.
- The demo mimics glasses input with on-screen buttons; a native display-glasses adapter is Planned.

7 Glasses app

7.1 Overview

Field operators walk rounds with gloves on, a radio in one hand and a log sheet in the other. They need the next inspection, the live reading at the asset in front of them, the nearby alarm and the right procedure without unlocking a phone or walking back to the control room. The glasses app is one web viewport (`apps/web/src/glasses`, served at `/glasses`) that puts this work at eye level. It records only: Hydris never operates equipment, and acknowledging an alarm says so.

Its principles are enforced in `hud.tsx`: one primary action per screen and six directional keys or voice for everything (NFR-021); body text of 20 px or more with light strokes on black for additive displays (NFR-020); cached data labelled with its age, never as live (NFR-026); and what a device shows decided in the console, not on the device (FR-054).

7.2 Users and user stories

Table 20. Glasses app user stories

ID	Story	Priority
US-G1	As a field operator I want to record round steps hands-free so that I do not carry a log sheet	P0
US-G2	As a field operator I want steps saved when the network drops so that no work is lost	P0
US-G3	As a field operator I want to acknowledge a nearby alarm and open its SOP so that I act on the right procedure	P0
US-G4	As a shift supervisor I want to pair glasses from my phone so that no password is typed on the device	P0
US-G5	As an admin I want to change what a role sees without an app update so that screens fit the job	P1
US-G6	As a prospect I want to run the full app in a simulator so that I can evaluate it without hardware	P1

7.3 Scope

Table 21. Glasses app scope

In scope (Built)	Out of scope or Planned
HUD layout, phone layout for the bridge, browser simulator	Rendering on a real glasses display (Planned)
Pairing by code and QR, device tokens, log out and unpair	Native Vuzix or RealWear hosts (Planned)
Home, alerts, inspections with offline queue, tasks, measurements, assets, map	Photo steps from the glasses camera (FR-045 Partial)
SOP library and reader with offline cache	Hardware gestures such as a temple touchpad (Planned)
Audit scan and Hey Hydris entry points	Multi-language voice (FR-073 Planned)

7.4 Runtimes

One React viewport runs in three places. `useViewportLayout` chooses phone mode inside the Android bridge (`HydrisNative` present), with `?mode=phone`, or on a narrow portrait screen; otherwise HUD mode.

Table 22. HUD and phone runtimes

Aspect	HUD mode	Phone mode
Target	Display glasses, browser simulator	Bridge app on the phone paired with Ray-Ban Meta
Canvas	600 x 600, scaled to window	540-unit width, full height, safe areas
Navigation	Six keys, swipes, simulator pad	Bottom tabs (Home, Scan, Hey Hydris, Alerts) plus touch
Camera and voice	Simulated or device camera; browser speech	Glasses camera and microphone through the bridge

Twelve top-level screens (FR-059) sit under a Home menu whose order and visibility come from the UI profile; Home and Device can never be switched off. Navigation is a stack mirrored into browser history, so back never leaves the app.

7.5 Pairing

Pairing follows the shape of the OAuth device authorization grant (RFC 8628). The glasses show an 8-character code (no 0, O, 1, I, L or U) and a QR link valid for 300 s; a signed-in supervisor approves on `/pair`; the glasses poll every 3 s with a secret that is never displayed and receive a 15-minute access token and a 30-day rotating refresh token. On boot, a revoked, suspended or unassigned device shows that state plainly. Log out and unpair, behind a confirmation, ends the device sessions (FR-058).

7.6 Inspections with offline replay

The runner walks one step at a time through check, reading, choice and text steps; out-of-range readings, failed checks and abnormal choices are marked abnormal automatically. Every response is queued with a client operation ID before it is sent, so online and offline work share one path. The status bar shows "N queued"; on reconnect the queue replays once, idempotently, and the app reports the synced count or names the first conflict. Submitting needs a connection.

7.7 UI profiles

A profile is a versioned description of enabled screens and order, home widgets, allowed actions, a measurement filter and a text scale (0.9, 1, 1.15 or 1.3, NFR-019). Precedence is device, then user, then role, then organization. Publishing sends `profile_changed`, the glasses reload context without a restart, and the API enforces the same profile with `403 profile_forbidden`. A blocked action shows "not permitted by your profile" rather than disappearing.

7.8 Functional requirements

Table 23. Glasses app requirements

ID	Requirement	Pri	Status
FR-059	Heads-up app, 12 screens, six-key input, phone layout	P0	Built
FR-050	Pairing by code and QR, 300 s TTL, 3 s polling	P0	Built
FR-051	Device tokens: 15-minute access, 30-day rotating refresh	P0	Built
FR-052	Suspend, revoke, reassign; effective within 30 s	P0	Built
FR-058	Log out and unpair from the glasses menu	P0	Built
FR-054	UI profiles: draft, publish, rollback, live push	P1	Built
FR-041	Inspection execution with offline queue and replay	P0	Built
FR-042	Submit inspections; returned ones can restart	P0	Built
FR-045	Photo steps from the glasses camera	P1	Partial
FR-112	Alarm detail opens the linked SOP	P0	Built
FR-113	Opened SOPs cached for offline reading	P1	Built

ID	Requirement	Pri	Status
FR-060	Guided A4 capture with auto-capture	P0	Built
FR-071	Hey Hydris wake phrase, short spoken answers	P0	Built
NFR-025	Inspection responses survive connectivity loss	P0	Built
NFR-026	Cached context shown as cached, never live	P1	Built

7.9 Key flows

Every step is queued first, so a dropped network never loses work.

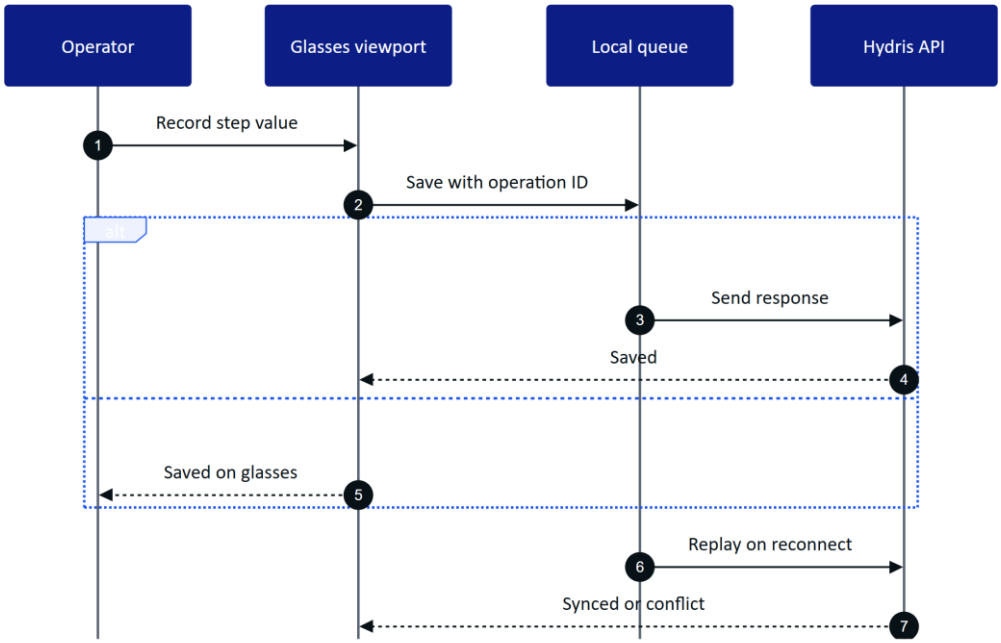


Figure 11. *Recording a step online or offline (data.ts)*

The viewport passes through explicit phases so device state is never hidden.

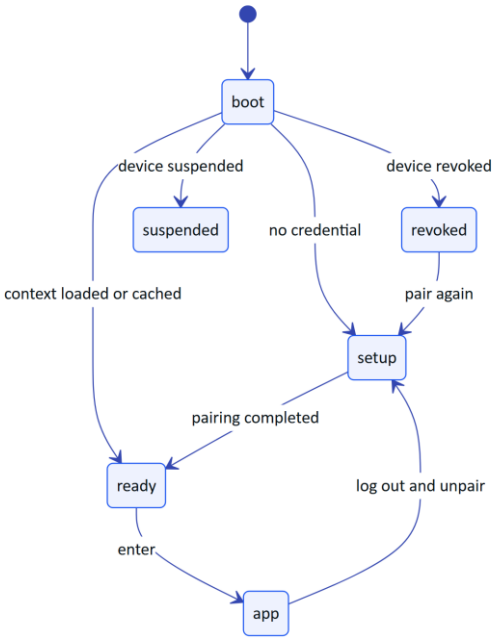


Figure 12. *Viewport phases from boot to the app*

7.10 Acceptance criteria

- Given an operator offline mid-round, when steps are recorded and the network returns, then every queued step syncs once with zero lost or duplicated (FR-041, NFR-025).
- Given a pairing code older than 300 s, when a supervisor tries to approve it, then approval is refused (FR-050).
- Given a profile that hides alarms, when the device calls the alarms API, then it receives `403 profile_forbidden` (FR-054).
- Given a profile is published, when the glasses are on, then the menu changes without a reload (FR-054).
- Given an alarm detail, when the operator acknowledges, then the app confirms that equipment state is unchanged (FR-059).

7.11 Success metrics

Table 24. Glasses app metrics (targets, not measured)

Metric	Target
Home ready after wake	2 s from cache, fresh data within 5 s (NFR-005)
Queued steps lost after reconnect	Zero (Playwright journey passes today)
Voice answer, end of speech to answer	3 s rules engine, 8 s model (NFR-003)
Rounds completed on glasses in pilot	Set with the pilot customer

7.12 Dependencies and open issues

- Not yet run on a real glasses display; Vuzix M400, RealWear Navigator 520 and Meta Ray-Ban Display are Planned targets.
- The refresh token sits in local storage until hardware-backed storage exists.
- Depends on the console for pairing approval and profiles, and on the Ray-Ban Meta bridge for phone mode hardware.

8 Hydris Field Android app

8.1 Overview

Operators in effluent and sewage treatment plants walk rounds with a phone in their pocket and, increasingly, Ray-Ban Meta glasses on their face. The only app that talked to those glasses was Meta AI, a consumer assistant that knows nothing about the plant. Hydris Field (`apps/android`, Kotlin and Jetpack Compose) is the native companion for the working day: one Hydris account, the plant dashboard, alarms and work, audit scanning, the Hey Hydris assistant and the glasses link in one app. Meta AI stays installed as the connection broker, because Meta's Wearables Device Access Toolkit (DAT) requires it for pairing and permission prompts; after a one-time setup the operator opens only Hydris Field.

8.2 Users and user stories

Table 25. Android user stories

ID	User story	Priority
US-A1	As a field operator I want to see what needs attention on my sites so that I act on the right thing first	P0
US-A2	As a field operator I want to acknowledge and resolve an alarm where I stand so that the console reflects reality at once	P0
US-A3	As a field operator I want to scan a paper log with my phone or glasses so that it reaches review without retyping	P0
US-A4	As a maintenance engineer I want to tick checklist items in the field so that work orders stay current	P1
US-A5	As a shift supervisor I want plant status, alarm trend and asset health in my pocket so that I can supervise while walking	P1
US-A6	As a prospect I want a named demo account so that I can try the full product in minutes	P0

All users already hold a Hydris console account. The app adds no new account type and needs no mobile-specific server authentication.

8.3 Scope

Table 26. Android scope

In scope	Out of scope
Sign-in, demo mode, session restore	Administration of members, roles, sites, rules (console)
Home dashboard, five Work tabs, alarm and sensor detail	Reviewing and approving audit documents (console)
Audit scan from glasses, phone camera or import	Equipment control of any kind
Hey Hydris assistant and the Glasses tab	A heads-up display on Ray-Ban Meta (they have none)
Realtime alarm notifications while the app runs	iOS native app; replacing Meta AI

The app has five bottom tabs (Home, Work, Scan, Hydris, Glasses) and two detail screens (alarm, sensor). Each tab keeps its scroll position. Every glasses feature has a phone fallback: phone camera, phone microphone, phone speaker. Permissions are asked in context and the app keeps working when one is denied.

8.4 Functional requirements

Table 27. Android functional requirements

ID	Requirement	Priority	Status
FR-089	Home and console overview from the same endpoint	P0	Built
FR-090	Console sign-in; session restored from a Keystore-encrypted refresh cookie	P0	Built
FR-091	Demo sign-in with a name of two or more characters	P0	Built
FR-092	Home: greeting, KPI tiles, 14-day trend, asset health, live measurements with freshness	P0	Built
FR-093	Work tabs (alarms, inspections, maintenance, sensors, audit docs) scoped to the user's sites	P0	Built
FR-094	Alarm detail with lifecycle actions; resolve requires a note	P0	Partial
FR-095	Sensor detail with 144-point history and typical range	P1	Built
FR-096	Audit scan from glasses, phone or import	P0	Built (phone); glasses Partial
FR-097	Realtime notifications for critical and high alarms while the app runs	P0	Built
FR-098	Offline outbox with idempotent replay	P1	Planned (GA)
FR-099	Complete MFA sign-in in the app	P1	Planned (Pilot)
FR-044	Run an inspection on the phone	P1	Planned

FR-094 is Partial because buttons are not yet driven by the API's list of allowed actions: a button may be offered that the server then refuses. Server rules always win, and the screen reloads the true state with the server's message.

8.5 Key flows

Session restore keeps the operator signed in across app restarts without a password: the short-lived access token lives in memory, and the rotating refresh cookie is stored with AES-256-GCM under a non-exportable Android Keystore key (`SecureStore.kt`).

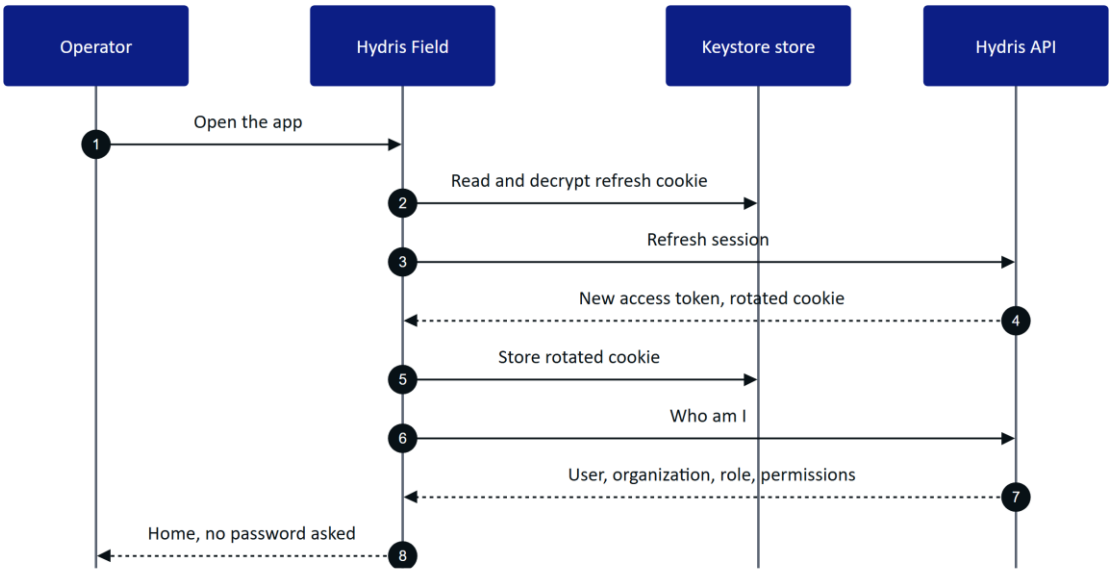


Figure 13. *Session restore when the app starts*

Alarms reach the phone over the server-sent event stream, which resumes with `Last-Event-ID` after a network drop; visible screens also poll (Home and alarms every 15 s, sensors every 10 s, audit documents every 20 s).

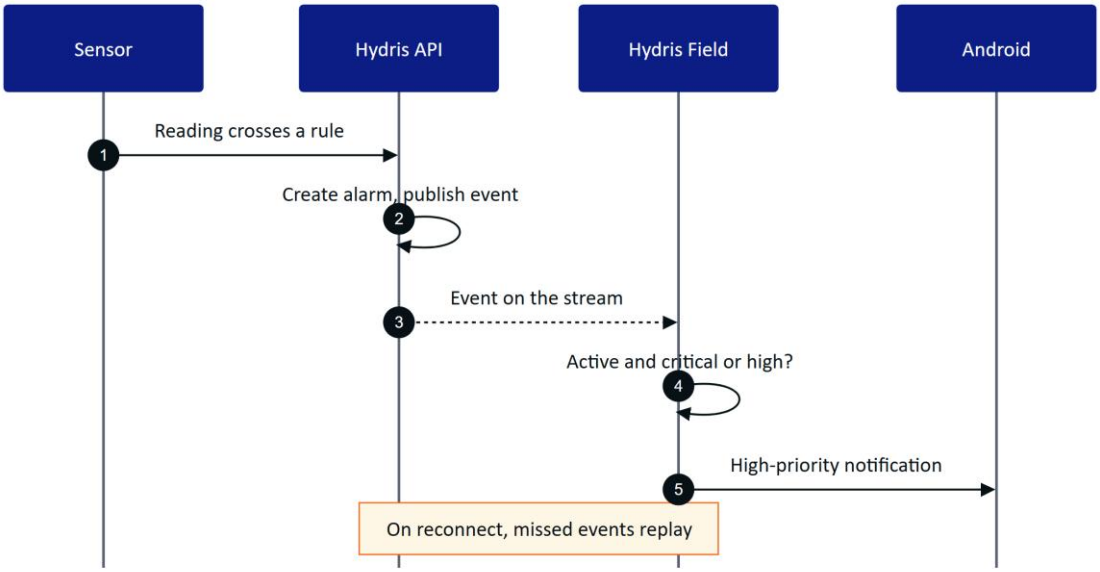


Figure 14. *Realtime alarm notification from reading to phone*

> **DECISION**
> Polling plus the event stream is deliberate for the pilot: every screen stays correct after a network drop without an offline cache. The cost is extra requests while a screen is visible. The Planned outbox and cache (FR-098) change this.

8.6 Acceptance criteria

- **FR-090:** Given a signed-in operator who kills the app, when they reopen it, then Home loads without a password; given a refused cookie, then Login shows "Session ended. Please sign in again."
- **FR-090:** Given several screens hit a 401 at once, when the app refreshes, then refreshes are serialized and only one rotated cookie is presented, so the server never sees reuse.
- **FR-094:** Given an acknowledged alarm, when the operator taps Resolve with an empty note, then the button stays disabled; with a note, then the alarm resolves and the console sees the update.
- **FR-097:** Given the app is running, when a critical alarm is created on the operator's site, then a high-priority Android notification appears.
- **FR-092:** Given a sensor with no readings, when Home renders, then it shows "No data", never zero, and synthetic values carry a "synthetic" label.
- **FR-096:** Given a printed lab sheet imported on the phone, when it is sent, then values are extracted and uploaded for review (verified manually on an emulated Pixel 7, API 35: 10 values including pH 7.3).

8.7 Success metrics

Table 28. Android success targets (targets, not measured results)

Measure	Target
Opening Scan to "Sent" for one A4 page	Under 60 s on a mid-range phone
Scans that auto-capture without "Capture now"	Above 70 %
Critical alarm to phone notification	Under 5 s while the app runs
Spoken answer after the question ends	Under 3 s on 4G

Measure	Target
Crash-free sessions in pilot	Above 99.5 %

8.8 Dependencies and open issues

- **Dependencies:** the Hydris API and its event stream; Meta AI app and DAT for glasses; Android 12 or later (`minSdk` 31, NFR-029).
- **Pilot gates:** MFA in the app (FR-099), release signing (builds are signed with the debug key and may not leave the team), and the first glasses hardware run (Q4 2026).
- **Notifications** work only while the app runs; FCM push is Planned for GA (FR-034), and a notification opens the app rather than the alarm.
- **Quality:** no automated Android tests yet; unit and UI tests are required before pilot.
- **Permissions:** Bluetooth and notification prompts repeat on every start; asking once is Planned.
- **Multi-site:** glasses are registered to the first site; a site picker is needed before multi-site pilots.

> **RISK**
> Until release signing is in place no build may leave the team. MFA, release signing and the first hardware run gate the pilot handover.

9 Ray-Ban Meta integration

9.1 Overview

Ray-Ban Meta glasses are the lightest way to give an operator a camera, microphone and speakers at eye and ear level, but they have no display. Hydris uses them as hands-free sensors for the phone: the glasses camera frames and shoots paper logs, the microphone hears "Hey Hydris", the open-ear speakers play the answer, and the capture button takes the photo. The phone is the screen. Integration uses Meta's Wearables Device Access Toolkit (DAT 1.0.1) with core, camera, speech and inputs modules.

> **IMPORTANT**
> No Hydris build has yet run against physical Meta glasses. DAT calls compile and type-check, and the bridge page was tested with a stand-in. Hardware behaviour is Built, not hardware-verified, and requirements below are Partial until it is.

9.2 Users and user stories

Table 29. Ray-Ban Meta user stories

ID	Story	Priority
US-R1	As a field operator I want to scan a paper log with the glasses camera so that my hands stay free	P0
US-R2	As a field operator I want spoken answers on the glasses speakers so that I keep my eyes on the plant	P0
US-R3	As a supervisor I want glasses registered as revocable devices so that I know who captured each scan	P0
US-R4	As an operator without glasses I want phone fallbacks so that I lose nothing but convenience	P0
US-R5	As a solutions engineer I want a clear setup path so that pilot kits work first time	P1

9.3 Scope

Table 30. Ray-Ban Meta scope

In scope	Out of scope
Ray-Ban Meta Gen 1 and Gen 2 (primary Gen 2)	Video recording or upload
Camera preview for guided capture, full-resolution photo	Replacing the Meta AI app
Glasses microphone, phone TTS on glasses speakers	Hydris UI on the glasses (no display)
Capture button short press	Long press and other gestures
Registering glasses as a Hydris device	Several glasses per phone

9.4 Meta AI app dependency and DAT lifecycle

Meta requires its Meta AI app to pair the glasses to the phone, update firmware, enable Developer Mode for unpublished apps, and approve each third-party app and each permission.

> **DECISION**
> The Meta AI app must stay installed and signed in. Hydris checks for package `com.facebook.stella` and, if missing, tells the operator to install it and pair there first. Hydris pairing is separate: it makes the glasses a managed device in the customer's organization.

DAT initializes only after Android grants Bluetooth permissions. Registration hands off to the Meta AI app and returns through a callback URL. Camera and microphone are DAT permissions approved in the Meta AI app, requested one at a time; a cancelled prompt counts as denied. One device session is shared by camera, speech and inputs: created when first needed, reused while started, and rebuilt after it stops. Preview frames (medium quality, 24 fps, at most every 100 ms to detection) are processed in memory for framing only and never stored; the photo is rotated upright and flattened to A4 (1654 x 2339). Images leave the phone only on "Send for verification".

Two routes share DAT and the backend: Hydris Field, the native Compose app and pilot runtime, and the Glasses bridge, a WebView hosting the `/glasses` viewport in phone mode for development comparison.

9.5 Registration

The Meta AI app pairs glasses to the phone; Hydris registration makes them a managed device assigned to a person and site, visible and revocable in the console, and credited for scans (FR-057). In the native app the signed-in user self-approves the device flow. The hardware ID is `rbm-` plus a SHA-256 digest of the DAT identifier, so the raw Meta ID never leaves the phone and re-registering reuses the same record. Glasses owned by another organization return `device_owned_elsewhere`.

9.6 Functional requirements

Table 31. **Ray-Ban Meta requirements**

ID	Requirement	Pri	Status
FR-055	Glasses camera via DAT: preview and full-resolution photo	P0	Partial (not on hardware)
FR-056	Capture button takes the photo or starts a scan	P1	Partial (not on hardware)
FR-057	Android app registers glasses as a Hydris device	P0	Partial (not on hardware)
FR-074	Glasses microphone with phone fallback, answers on speakers	P0	Partial (phone path Built)
FR-077	"Hey Meta, start Hydris" opens the assistant	P2	Partial (bridge only)
NFR-015	Camera only on explicit intent, no continuous recording	P0	Built
NFR-028	4-hour shift with up to 20 scans	P1	Planned (measure)
NFR-029	Android 12 or later	P0	Built

9.7 Key flows

Registration with Meta happens once, in the Meta AI app, before any Hydris capability is used.

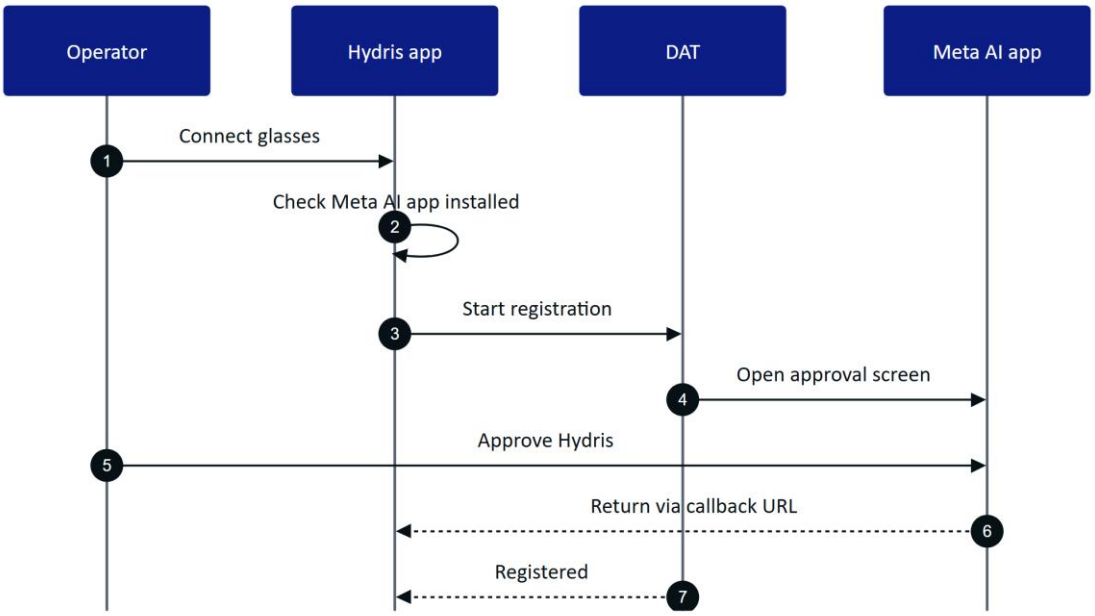


Figure 15. *DAT registration hands off to the Meta AI app*

One shared session serves camera, speech and inputs and is rebuilt cleanly after the glasses go away.

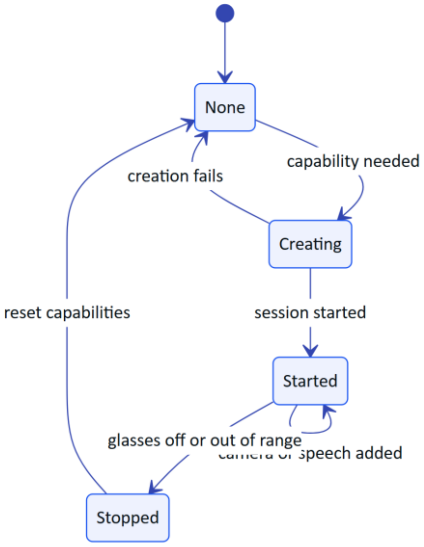


Figure 16. *DAT device session lifecycle*

9.8 Acceptance criteria

- Given the Meta AI app is missing, when the operator taps Connect glasses, then Hydris explains how to install and pair it (decision above).
- Given an active glasses preview, when the page is aligned and the capture button is pressed, then a full-resolution upright A4 image is returned (FR-055, FR-056).
- Given registered glasses, when an admin revokes them in the console, then later uploads from that device are refused (FR-057, GLS-T12).
- Given no glasses session, when the operator scans or asks a question, then the phone camera and microphone are used instead (FR-074).

9.9 Success metrics

Table 32. Hardware run targets (not measured)

Path	Target
First preview frame with an existing session	Under 3 s
Preview rate reaching detection	8 to 10 fps
Photo returned after capture	Under 4 s
Speech restart gap after silence	Under 1 s
Battery	4-hour shift with up to 20 scans

9.10 Dependencies and open issues

- Fifteen hardware acceptance tests (GLS-T01 to T15) are defined and none has run; first run on Gen 2 is the next step.
- The DAT camera stream is stopped only when the session ends; battery and capture-LED impact to be checked.
- Public DAT publishing is not open; pilots rely on Developer Mode and Developer Center release channels.
- Bridge hardening (navigation and permission restriction) is Planned before any distribution outside the team.
- The same glasses paired through both routes create two device records; pilots use one runtime per pair. A site picker for registration is Planned.

10 Audit document scanning

10.1 Overview

Treatment plants still run on paper: shift logbooks, round sheets, lab reports, permits and calibration certificates. These records are the evidence a regulator or auditor asks for, yet they are hard to search, easy to lose and slow to copy by hand, where a moved decimal point turns pH 7.1 into 71. Audit scanning turns a paper record into a verified, searchable PDF. The governing rule is dual verification: a machine cross-check of every value between independent readings, then a human review of every value and an independent approval by a second person.

10.2 Users and user stories

Table 33. Audit scanning user stories

ID	User story	Priority
US-S1	As a field operator I want to capture a paper record with glasses or phone in under a minute so that evidence is digital at the point of work	P0
US-S2	As a reviewer I want to confirm, correct or flag every value against the paper so that nothing wrong is verified	P0
US-S3	As a plant manager I want to approve or reject a verified document so that a second person stands behind it	P0
US-S4	As a compliance auditor I want a searchable PDF with the full verification trail so that I can rely on the record	P0
US-S5	As a reviewer I want to bulk-confirm values where engines agree confidently so that review is fast	P1

10.3 Scope

Table 34. Audit scanning scope

In scope	Out of scope
Capture from glasses, phone camera, imported photo, console	Signing on the device: devices capture, people verify
Guided A4 framing, auto-capture, flattening	Electronic signature with re-authentication (Planned, FR-068)
Two on-device OCR passes; fields, tables, figures	Handwriting-specific recognition models
Plausibility checks and an optional vision engine	Bulk upload of existing PDF archives
Field review, verify, approve, PDF with trail	Writing values into plant records automatically

10.4 Functional requirements

Table 35. Audit scanning functional requirements

ID	Requirement	Priority	Status
FR-060	Guided A4 capture with live detection, auto-capture, perspective correction	P0	Built (web and Android)
FR-061	Dual-pass on-device OCR (original and enhanced)	P0	Built
FR-062	Fields, tables, figures with plausibility checks	P0	Built

ID	Requirement	Priority	Status
FR-063	Optional vision engine, each value cross-checked against OCR in its box	P1	Built (needs a key)
FR-064	Field review; original value kept; verify needs zero pending and flagged	P0	Built
FR-065	Independent approval by a different person (four eyes)	P0	Built
FR-066	PDF with trail, searchable original and reconstructed page	P0	Built
FR-067	Up to 6 pages, each at most 900 KB, JPEG or PNG	P0	Built
FR-068	Electronic signature with re-authentication	P1	Planned (Advanced)
FR-069	Bulk-confirm values where two engines agree with confidence 0.8 or more	P1	Built
FR-096	Scan from glasses, phone or import in the Android app	P0	Built (phone); glasses Partial
FR-055	Glasses camera capture through DAT	P0	Partial (not run on hardware)

Guided capture speaks or shows one instruction at a time, in a fixed order (find page, move back, move closer, straighten, hold still), checking frames at most every 90 ms. It auto-captures after 900 ms of readiness and flattens the page to A4 at 200 dpi (1654 x 2339 pixels). Phone, glasses viewport and console share the same detection code and thresholds.

10.5 Key flows

The journey has three halves that never mix: the device captures and reads, the server structures and cross-checks, and people verify and approve.

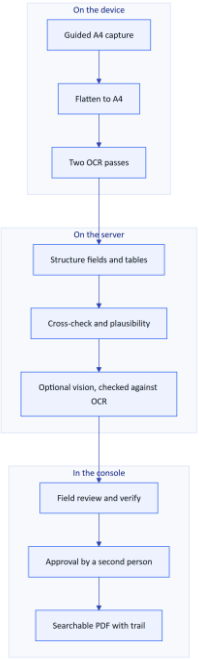


Figure 17. *Paper record to verified PDF*

Every review action is appended to the document's event history; any change after verification sends the document back to review, and approval must come from a different person.

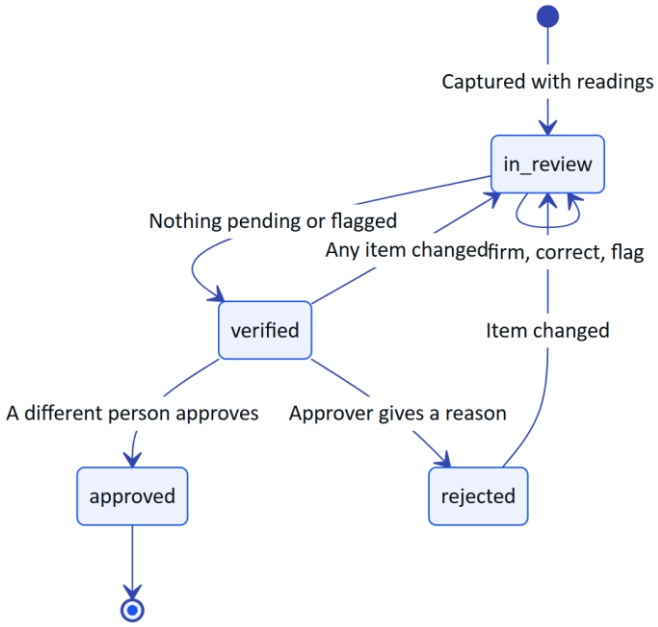


Figure 18. *Audit document lifecycle and the four-eyes rule*

Each value carries an agreement level between the passes (match, close, mismatch) and keeps its position on the page. Plausibility checks run even when both passes agree, because both can be wrong the same way.

> **DECISION**
> The primary quality metric is the silent error rate: a value marked as agreed and not flagged, but wrong. A system that is 85 % accurate and flags what it is unsure of is safer than one that is 95 % accurate and confidently wrong on the rest.

Table 36. Safeguards for the numbers that matter

Risk	Safeguard
Decimal point dropped or moved	"7.4" against "74" is always a mismatch; thousands separators are not
Both passes misread the same way	Range checks flag pH 71 and offer a matching labelled value as the fix
Letters and digits confused	O and 0 are folded inside numbers before comparison
Vision model invents a value	Checked against OCR text inside the value's box
Reviewer overwrites the machine value	The first extracted value is kept beside the correction

10.6 Acceptance criteria

- **FR-060:** Given an A4 sheet held aligned and still, when it is ready for 900 ms, then the device captures and flattens it without a button press.
- **FR-062:** Given both passes read pH as 71, when the server structures the page, then the value is flagged as implausible with a reason and a suggested fix.
- **FR-063:** Given a vision value with no supporting OCR text in its box, when results merge, then it is never marked as agreed.
- **FR-064:** Given a document with one pending or flagged item, when the reviewer tries to verify, then verify is blocked.
- **FR-065:** Given a verified document in a customer organization, when the verifier tries to approve it, then approval is refused; demo tenants allow self-approval and record it.

- **NFR-031:** Given two reviewers editing the same document, when the second saves with an old revision, then the write returns `409 stale` instead of overwriting.

10.7 Success metrics

Table 37. Scanning quality targets (targets, not measured results)

Metric	Pilot target
Fields where the engines agree, typed forms	90 % or more
Fields a reviewer corrects	Under 5 %
Silent error rate	Under 0.5 %
Decimal correctness	99 %
Planted implausible values flagged	100 %
Capture to review-ready per page	15 s or less
Capture to approval, median	Under 24 hours

One manual demo (emulated Pixel 7, API 35) read a printed lab sheet correctly, including pH 7.3, with 10 values extracted. That is one sheet, not a benchmark.

10.8 Dependencies and open issues

- **Benchmark not yet run.** A 120-page ground-truth set is designed; until it runs, every accuracy number is a target.
- **Glasses capture unproven on hardware.** The DAT path works on Meta's mock device; the first Ray-Ban Meta run is scheduled for Q4 2026.
- **Vision consent.** A tenant must agree to vision processing and the deployment must record where inference runs; until then pilots run on OCR alone.
- **Storage.** Page images move to object storage during Hardening.
- **Electronic signature** (FR-068) is Planned for the Advanced phase.

11 Hey Hydris voice assistant

11.1 Overview

An operator at a clarifier with gloves on cannot scroll a dashboard. Hey Hydris lets them ask out loud ("what's the outlet pH?", "any open alarms?", "what are my tasks?") and hear a short answer on the glasses speakers or the phone, with the facts behind it on screen. Every answer comes from the caller's own plant data through read-only, tenant-scoped tools, never from general knowledge. The assistant is advisory only and cannot change anything.

11.2 Users and user stories

Table 38. Voice assistant user stories

ID	User story	Priority
US-V1	As a field operator I want a reading, its age and alarm state by voice so that I keep my hands on the job	P0
US-V2	As a maintenance engineer I want to hear my open work and an asset's procedure so that I act without the console	P0
US-V3	As a supervisor I want to hear open alarms on my sites while walking so that I stay aware	P1
US-V4	As any user I want to ask a follow-up without repeating context so that the conversation feels natural	P1

11.3 Scope

Table 39. Voice assistant scope

In scope	Out of scope
Wake phrase, spoken question, spoken answer	Operating equipment or changing records
Readings, alarms, my work, asset status, approved SOPs	Answers from outside the tenant's data
Follow-ups with history (up to 8 turns)	Indian languages (Planned, FR-073)
Glasses, phone and browser speech	"Hey Meta, start Hydris" as a supported path (experimental)

11.4 Functional requirements

Table 40. Voice assistant functional requirements

ID	Requirement	Priority	Status
FR-070	Grounded SOP retrieval from approved documents with a safety notice	P0	Built
FR-071	Wake phrase and short spoken answers	P0	Built
FR-072	Answers only from tenant data through scoped tools; rules fallback never guesses	P0	Built
FR-073	Hindi, Tamil, Marathi, Gujarati	P1	Planned (Advanced)
FR-074	Glasses microphone with phone fallback; answers on glasses speakers	P0	Partial (not run on hardware)
FR-075	Facts with source; synthetic data labelled	P0	Built
FR-076	Follow-ups with history (up to 8 turns)	P1	Built

ID	Requirement	Priority	Status
FR-077	"Hey Meta, start Hydris" opens the assistant	P2	Partial (experimental)
FR-078	No write tools	P0	Built

11.5 Key flows

When an Anthropic credential is configured, Claude answers with up to five rounds of five read-only tools (readings, open alarms, my work, asset, procedures) that run in the caller's tenant and site scope; otherwise, or on any model error, a deterministic rules engine answers.

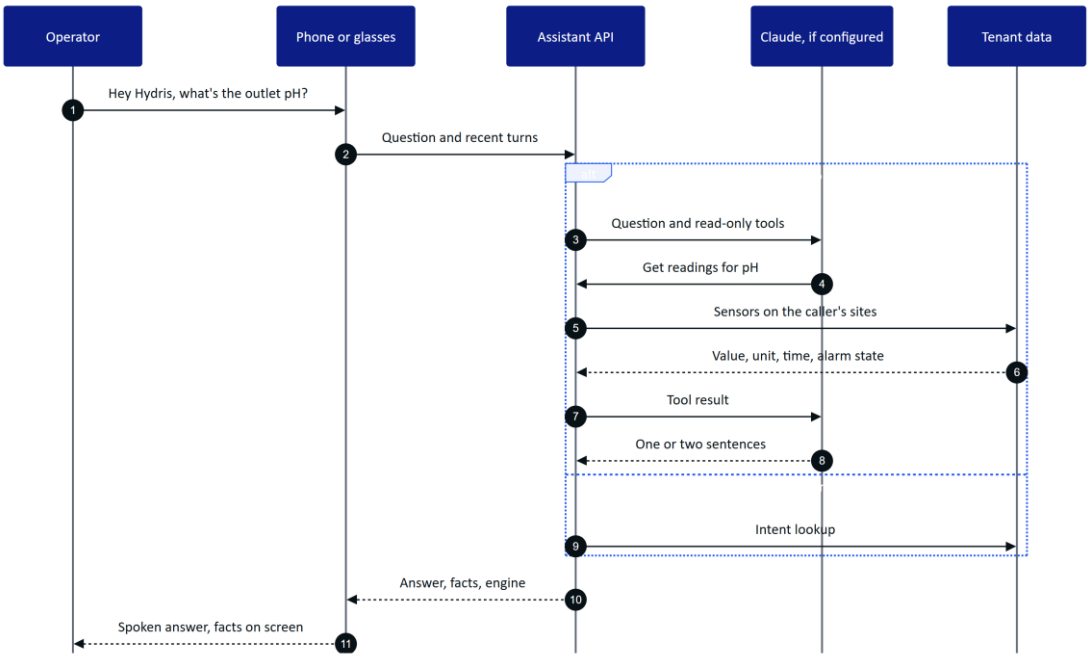


Figure 19. *A spoken question answered from the caller's own data*

The wake phrase is detected on the device with a pattern tolerant of how recognizers hear "Hydris". Transcripts are dropped while the assistant speaks, so it never answers itself; "stop", "cancel" or "thanks" ends the follow-up window.

11.6 Acceptance criteria

- **FR-072:** Given no model key, when the operator asks about the weather, then the answer is "I can't answer that from your plant data".
- **FR-072:** Given a site-scoped operator, when they ask about another site's sensor, then the answer contains nothing they could not see in the console.
- **FR-075, NFR-036:** Given a reading 12 minutes old, when it is answered, then the answer states its age; a stale reading adds "not current, check the sensor"; missing data is "no data", never zero.
- **FR-078:** Given any question, when the assistant runs, then only read tools are available and no record changes.
- **FR-074:** Given glasses linked but the glasses microphone denied, when the assistant listens, then it falls back to the phone microphone, then to typing.

11.7 Success metrics

Table 41. Voice targets (targets, not measured results)

Metric	Pilot target
Questions answered with facts, not "can't answer"	85 % or more

Metric	Pilot target
Numbers in answers missing from facts	0
Spoken answer after the question ends	Under 3 s (rules) and 8 s (model) on 4G, NFR-003

11.8 Dependencies and open issues

- **Logging** of engine and intent is Planned for the pilot; until then the targets cannot be measured.
- **Glasses microphone** on Ray-Ban Meta hardware is a first hardware run item for Q4 2026.
- **False accepts** of the wake phrase are tested against eight hours of plant conversation before the pattern is tightened.
- **Follow-up window:** the glasses viewport closes after 8 s of silence; Android waits for a stop phrase. Aligning on 8 s is Planned.
- **Model consent:** a tenant must agree to Claude processing; otherwise the rules engine runs alone.

12 Platform, data and integrations

Every Hydris surface (website, operator console, glasses HUD, Hydris Field Android app) talks to one platform: a modular API over one Postgres schema with row-level security. This section states what the platform must guarantee to every product: isolation, security, honest data, safe integration and AI that never acts without a person.

12.1 Architecture summary

Hydris is an npm workspaces monorepo. The API (`apps/api`, Fastify 5) is a modular monolith of fifteen route modules sharing core services (tenant context, event bus, audit, storage, crypto). It runs inside the Next.js web deployment through a catch-all route that calls `app.inject` (no network hop), or as its own container. A pure TypeScript domain package (`packages/domain`) holds state machines, permissions, freshness and voice parsing rules shared by server and web; the Android app mirrors them by hand.



Figure 20. *Platform containers: all clients share one API and one event stream*

One validated configuration (`apps/api/src/config.ts`) selects Postgres or in-process PGlite, the realtime mode, cloud or disk storage, and an external API. The same code runs on Vercel, in Docker Compose at a plant, in a hybrid set-up and as the self-contained public demo.

> **DECISION**
> A modular monolith (ADR-002) with the API inside the web deployment (ADR-003) gives one deployable and in-process calls for the pilot. The trade-off is module discipline and a cold start that builds the app and migrates.

12.2 Tenancy and security requirements

Isolation is enforced by Postgres, not only by application code. Every tenant table has row-level security enabled and forced, with one policy comparing the row's organization to a per-transaction setting (`SET LOCAL`, so nothing leaks across pooled connections). Requests run as the non-privileged role `hydris_app`. On top sit RBAC (six roles, 38 permissions in `packages/domain/src/roles.ts`) and site scope; out-of-scope records return 404, not 403, so users cannot probe for other sites.

The principal is reloaded from the database on every request, so a revoked session or device stops at once. People sign in with argon2id-hashed passwords and optional TOTP MFA, receive a 15-minute access token and a 7-day rotating refresh token; reuse of a rotated token after a 30-second grace revokes the whole family. Devices never see a password: they

pair through an RFC 8628-style flow (8-character code, 300 s TTL, 3 s polling) and act for their assigned operator under a ceiling of 13 device permissions.

Table 42. Credentials

Credential	Lifetime	Revoked by
User access token	15 min (12 h in demo only)	Family revocation, checked per request
User refresh token	7 days, rotating	Logout, password change, reuse detection
Device access token	15 min	Session version bump on suspend, revoke, reassign
Device refresh token	30 days, rotating	Revoke, logout, reuse detection
Gateway key	Until revoked	Admin revoke; stored only as a hash

> **RISK**
> Owner and superuser connections bypass RLS. Today the same owner user runs migrations and the application; production must connect as a runtime member of `hydris_app` that is not the table owner.

12.3 Data entities summary

The schema (`apps/api/src/db/schema.ts`) has 42 tables in ten domains, each owned by one module. Plant objects are archived, not deleted. Records are evidence: readings keep provenance, corrections never overwrite the original, and synthetic data is refused for customer tenants by a database check.

Table 43. Core data entities

Domain	Key entities	Rule
Identity and tenancy	Users, organizations, memberships, membership sites, sessions, invitations	Users are global; membership and site scope are per organization
Plant	Sites, facilities, areas, assets, sensors, gateways, documents	Archive and restore, never delete
Telemetry	Readings, corrections	Readings immutable; one per sensor and time
Alarms	Rules, events	One open event per rule
Work	Inspection templates, routes, inspections, responses, maintenance tasks	Revision counter for offline replay
Devices and UI	Devices, assignments, device sessions, pairing challenges, UI profiles	One operator per device
Audit documents	Documents, pages, events	Append-only event trail
Platform	Audit log, realtime events, mail outbox	No update or delete on audit events

12.4 APIs and integrations

Hydris never connects into a plant. A customer-side gateway pushes readings outward over HTTPS (`POST /ingest/v1/readings`) with a hashed `hgw_` key, up to 500 readings per call, idempotent per sensor and timestamp, with per-reading rejections so the gateway can retry safely. Manual and lab entry and the demo simulator use the same `ingest()` path. Late data is stored for history but never overwrites a newer value or re-triggers rules; deadband and on-delay stop flapping.

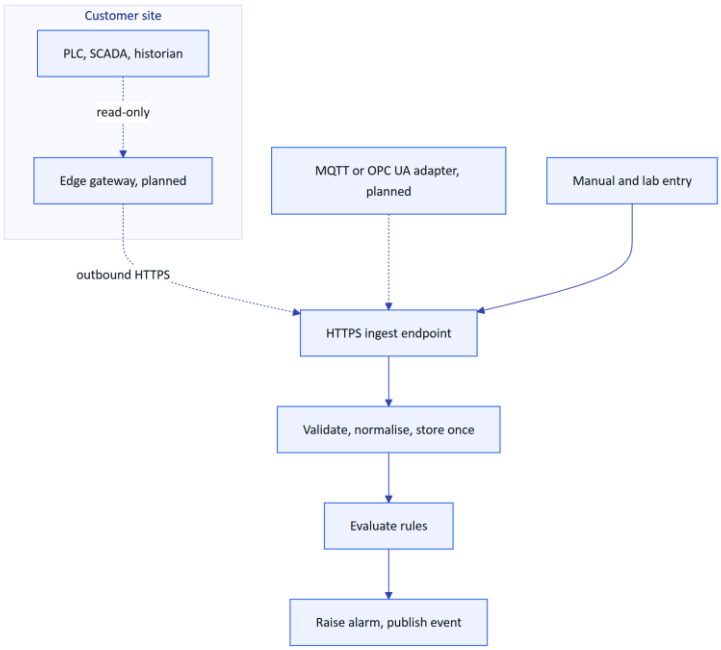


Figure 21. *Telemetry from the plant to an alarm; dashed paths are planned*

Realtime events (alarms, telemetry, work, devices) reach clients over Server-Sent Events opened with `fetch`, so the token travels in a header, never a URL. Each event is authorised per connection and carries only identifiers; clients refetch through the normal API. `REALTIME_MODE=memory` is immediate on a single instance; `db` mode fans out across instances with up to 1.5 s polling and replays from `Last-Event-ID`.

Table 44. **Integration status**

ID	Integration	Status
FR-022	HTTPS gateway ingestion, 500 readings per call, idempotent	Built
FR-023	Manual and lab entry with provenance	Built
FR-024	Append-only corrections	Built
FR-033	Realtime push of alarm events to all clients (SSE)	Built
FR-027	MQTT and OPC UA ingestion adapters	Planned (Advanced)
FR-028	Read-only SCADA and historian connectors	Planned (Advanced)
n/a	Edge gateway software (store and forward)	Planned; API contract Built

12.5 AI services with a human in the loop

Two features use Claude, and both are optional. Audit scanning adds a vision engine whose every value is checked against on-device OCR; the assistant calls five read-only tools inside the caller's tenant transaction, so RLS and site scope apply to every lookup. The model never sees the database. Without a key, scans keep their OCR results and the assistant uses a rules engine that never guesses. People decide what becomes a record: no document reaches verified with pending or flagged items, and approval needs a second person.

> **DECISION**
> AI adds reach, never unchecked trust (ADR-009, ADR-015). Every AI-extracted value passes human review, and the assistant has no write tools.

12.6 Platform requirements

Table 45. Platform requirements

ID	Requirement	Priority	Status
FR-003	Refresh rotation with family revocation on reuse	P0	Built
FR-007	Site-scoped permissions for every role except org admin	P0	Built
FR-008 / NFR-011	Database-enforced tenant isolation	P0	Built
FR-011	SAML 2.0 and OIDC single sign-on per organization	P1	Planned (GA)
FR-029 / NFR-030	Synthetic data labelled and refused for customer tenants	P0	Built
FR-050	Pairing with short code, QR, 5-min TTL, 3 s polling	P0	Built
FR-051	Device tokens with rotating refresh and reuse detection	P0	Built
NFR-001	API read latency p95 under 300 ms in-region (target)	P0	Partial (not measured)
NFR-002	Realtime alarm delivery 2 s or less (target)	P0	Built (design; not measured)
NFR-006	Availability 99.5 % in pilot, 99.9 % at GA (targets)	P0	Planned (measure)
NFR-007	Horizontal scale of realtime	P0	Built (db mode)
NFR-008	Scheduled jobs at a cadence suitable for real data	P0	Partial (daily on Vercel)
NFR-009	Daily backups, point-in-time recovery, quarterly restore test	P0	Planned
NFR-010	CI with typecheck and tests on every change	P0	Planned (Hardening)
NFR-013	Rate limiting (600 per min global, 10 per min sign-in)	P0	Built
NFR-014	Secrets at rest encrypted or hashed	P0	Built
NFR-017	Data residency in India for Indian customers	P1	Planned (GA)
NFR-018	Independent penetration test before GA	P0	Planned (GA)
NFR-034	Structured logs, error tracking, uptime alerts	P1	Planned (Pilot)
NFR-037	Graceful degradation without the model	P0	Built
NFR-039	Human in the loop for every AI-extracted value	P0	Built
NFR-040	Model provider and region documented per tenant	P1	Planned (GA)

RISK

On an all-Vercel topology, stale-data alarms run at most daily (NFR-008). A production plant needs a per-minute scheduler or the API in a container.

13 Non-functional requirements

Non-functional requirements apply across every product. IDs match the long PRD. Every number in the Target column is a target unless the Status column says it is built and verified by tests.

Table 46. Non-functional requirements by category

Category	ID	Requirement	Target	Status
Performance	NFR-001	API read latency	p95 under 300 ms in-region	Partial (not measured)
Performance	NFR-002	Realtime alarm delivery, creation to client	2 s or less	Built (design); not measured
Performance	NFR-003	Voice answer, end of speech to spoken answer	3 s rules engine, 8 s model engine	Partial (not measured)
Performance	NFR-004	OCR time per page on a mid-range phone	15 s or less for both passes	Partial (not measured)
Performance	NFR-005	Glasses home ready after wake	2 s cached, fresh data within 5 s	Planned (measure)
Availability	NFR-006	Service availability	99.5 % monthly in pilot, 99.9 % at GA	Planned (measure)
Availability	NFR-008	Scheduled jobs at a cadence fit for real data	Per-minute stale and pairing jobs	Partial (daily cron on Vercel)
Availability	NFR-009	Backups and restore	Daily backups, PITR, quarterly restore test	Planned
Availability	NFR-010	CI on every change	Typecheck, unit, API and e2e on every pull request	Planned (Hardening)
Security	NFR-011	Tenant isolation in the database	RLS on every tenant table; app role without BYPASSRLS	Built
Security	NFR-012	Short-lived credentials	15-minute access tokens; rotating refresh with reuse detection	Built
Security	NFR-013	Rate limiting	600 per minute global; 10 per minute sign-in	Built
Security	NFR-014	Secrets at rest	MFA secrets encrypted; tokens hashed; Android Keystore AES-256-GCM	Built
Security	NFR-018	Independent penetration test before GA	No open high or critical findings	Planned (GA)
Privacy	NFR-015	Camera only on explicit intent	No continuous recording or background capture	Built
Privacy	NFR-016	Images stored only for audit documents and attachments	No other path persists camera frames	Built
Privacy	NFR-017	Data residency in India for Indian customers	Database, storage and inference in an Indian region	Planned (GA)
Privacy	NFR-040	Model provider and region documented per tenant	Visible in tenant settings	Planned (GA)
Accessibility	NFR-019	Glasses text scale per profile	Scales 0.9 to 1.3	Built

Category	ID	Requirement	Target	Status
Accessibility	NFR-023	Console and Android meet WCAG 2.2 AA	Audit before GA	Partial (not audited)
Accessibility	NFR-024	Large touch targets for gloved use	Primary buttons 52 to 54 dp	Built
Hands-free	NFR-020	Additive-display friendly design	Pure black renders transparent; 600 by 600 viewport	Built (verify on hardware)
Hands-free	NFR-021	Every glasses action by six directional keys or voice	Up, down, left, right, select, back	Built
Hands-free	NFR-022	Spoken answers are short	One or two sentences	Built
Hands-free	NFR-028	Battery on Ray-Ban Meta	4-hour shift with up to 20 scans	Planned (measure)
Offline	NFR-025	Inspection responses survive connectivity loss	Queue with idempotent replay	Built
Offline	NFR-026	Last context and SOPs readable offline	Shown as cached, never as live	Built
Offline	NFR-027	Android offline outbox	Phone actions queue and replay	Planned (GA)
AI quality	NFR-035	Assistant uses only values from tools in the conversation	Zero numbers absent from tool results	Built (prompt and tool design)
AI quality	NFR-036	Assistant states reading age and stale or missing data	Every answer with a value	Built
AI quality	NFR-037	Graceful degradation without the model	Rules engine and OCR-only fallback	Built
AI quality	NFR-039	Human in the loop for every AI-extracted value	No verified document with pending or flagged items	Built

Table 47. AI and OCR quality targets for the pilot (targets, not measurements)

Measure	Target	How it will be measured
Fields where independent engines agree	90 % or more on typed forms	Agreement score; pilot sample of 200 pages
Fields a reviewer corrects	Under 5 %	Corrected over reviewed fields
Vision false confidence	Under 0.5 %	Blind re-check of 100 approved documents
Assistant answers with facts	85 % or more	Intent logging on `/assistant/ask`
Answers with a number not in the facts	0	Automated check on logged answers

NOTE

The security-critical paths are covered by 67 passing automated tests (26 domain, 41 API, local run on 22 October 2026) plus 4 Playwright journeys. Performance, availability and battery numbers have not yet been measured; measuring them is part of the pilot entry criteria.

14 Release plan and milestones

The MVP shipped between 14 and 22 October 2026: API, console, glasses app and simulator, pairing, UI profiles, audit scanning, voice, demo mode and the native Android app in code. The plan below sequences the rest. Dates are a plan, not a commitment, and depend on the dependencies and open questions in later sections.

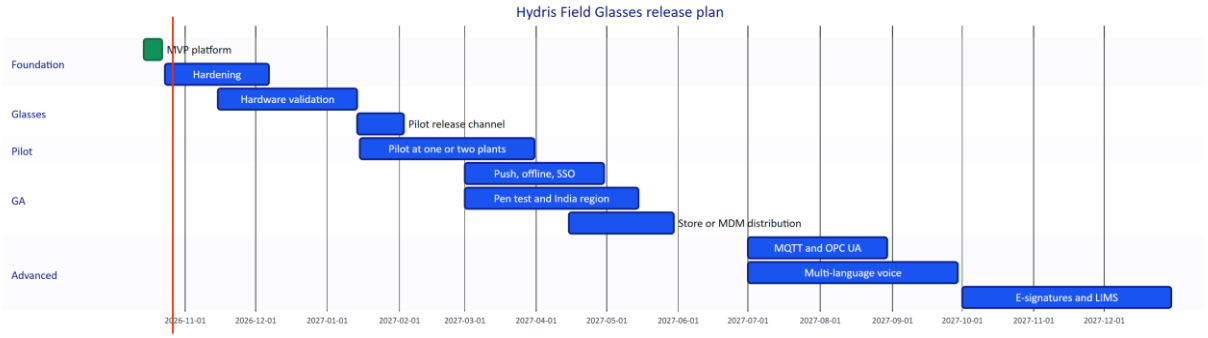


Figure 22. *Release plan, Q4 2026 to Q4 2027 (plan, indicative dates)*

14.1 Phase entry and exit criteria

Table 48. Release phases with entry and exit criteria

Phase	When (plan)	Entry criteria	Scope	Exit criteria
MVP	Done, October 2026	None	Platform core, all products in code	Live demo; 67 tests passing
Hardening	Q4 2026	MVP on `main`	Email (FR-086), CI (NFR-010), job cadence (FR-035, NFR-008), Android fixes (FR-094, FR-099), page images to object storage, observability (NFR-034)	All Must requirements for pilot Built or with an agreed workaround
Hardware validation	Q4 2026 to Q1 2027	DAT Developer Mode access; test glasses on hand	Camera, capture button, speech, registration (FR-055, FR-056, FR-057, FR-074); battery (NFR-028)	Every glasses requirement Built, or descoped with a phone fallback
Pilot	Q1 2027	Hardening exit; signed pilot contract; site camera policy agreed	One or two Indian ETPs or STPs, 10 to 30 operators, gateway ingestion, compliance summary (FR-120), escalation (FR-037), Android inspections (FR-044)	KPI baselines captured; targets met or explained; customer sign-off
GA	Q2 2027	Pilot exit	FCM push (FR-034), offline outbox (FR-098), SSO (FR-011), India region (NFR-017), pen test (NFR-018), recurring schedules (FR-047), Play or MDM	Store or MDM release approved; security review passed

Phase	When (plan)	Entry criteria	Scope	Exit criteria
Advanced	H2 2027 onwards	GA	MQTT and OPC UA (FR-027), SCADA connectors (FR-028), multi-language voice (FR-073), e-signatures (FR-068), native glasses display, LIMS, iOS	Driven by customer demand

14.2 Pilot exit criteria

The pilot is judged on evidence, not on feature count. It exits successfully when all of the following hold:

- ☐ KPI baselines are captured for paper rounds before go-live and for Hydris during the pilot.
- ☐ Pilot targets are met or each miss is explained with a remedy: 70 % weekly active operators, 95 % on-time rounds, under 5 minutes to acknowledge critical and high alarms, under 24 hours capture to approval.
- ☐ Zero hallucinated numbers in logged assistant answers and zero cross-tenant incidents.
- ☐ Every glasses feature used in the pilot runs on physical Ray-Ban Meta glasses or has a working phone fallback.
- ☐ Availability at or above the 99.5 % pilot target (NFR-006).
- ☐ The customer signs off and states whether the regulator accepts the Hydris PDF as a primary record (Q4).

DECISION

Every glasses feature must work on the phone first. This decouples the pilot from both hardware risk and Meta's distribution timeline.

15 Dependencies, assumptions and constraints

Table 49. Dependencies, assumptions and constraints

Type	Item	Impact if it fails	Fallback or response
Dependency	Meta AI app (`com.facebook.stella`) for pairing, registration, permissions and firmware	Glasses features unavailable	Phone camera and microphone paths
Dependency	Meta Wearables DAT 1.0.1 and Developer Mode	Cannot reach the glasses	Phone paths; Developer Mode for the pilot
Dependency	Wearables Developer Center registration	Cannot distribute beyond test devices	Apply early; pilot in Developer Mode
Dependency	Vercel hosting, cron and Blob storage	Hosted service outage; daily cron only	Container deployment (Docker, Render blueprint)
Dependency	Managed Postgres (Neon or similar)	Outage	Point-in-time recovery; PGlite for local and demo
Dependency	Anthropic API for assistant and vision	Degraded answers, no vision	Rules engine and OCR-only extraction (Built)
Dependency	Google ML Kit text recognition	Lower OCR quality on Android	Tesseract.js path in the web app
Dependency	Email provider (to be chosen)	Invitations and resets do not arrive	Admin copies links from `mail_outbox`
Assumption	Pilot plants have an HTTPS gateway or OCEMS analyzer that can push readings	No live telemetry	Manual and lab entry
Assumption	Operators carry an Android 12 or later phone (min SDK 31)	Android app cannot install	Glasses web app in phone layout
Assumption	Customers allow camera glasses in agreed zones	Scan disabled on site	UI profile disables Scan; phone scanning
Constraint	Read-only towards the plant	No control actions	By design (P1)
Constraint	English UI and voice until Advanced	Limited reach for non-English speakers	Multi-language voice Planned (FR-073)
Constraint	No iOS before GA	iPhone users excluded	Decision at GA planning (Q5)

16 Risks and mitigations

Table 50. Product risks with likelihood, impact and mitigation

ID	Risk	Likelihood	Impact	Mitigation	Owner
R1	Meta DAT publishing stays gated	Medium	High	Pilot in Developer Mode; phone paths ship independently; apply early	Product
R2	Meta AI app behaviour changes	Medium	Medium	Onboarding checklist; Glasses screen shows link state and connection log	Mobile
R3	Glasses paths fail on real hardware	Medium	High	Hardware matrix before pilot; phone fallback on every feature	Mobile, QA
R4	Email not wired	High	Medium	FR-086 in Hardening; interim admin copy of links	Platform
R5	Serverless cron runs once a day	High	High	Per-minute cron or container host; NFR-008	Platform
R6	No CI	Medium	Medium	NFR-010 in Hardening	Platform
R7	Android Resolve fails without a note	High	Medium	FR-094 fix in Hardening	Mobile
R8	LLM availability, cost or residency objections	Medium	High	Rules and OCR-only modes; region-pinned provider; NFR-040	Product
R9	Camera glasses not accepted on site; bystander privacy	Medium	Medium	Camera on intent only; UI profile can disable Scan; site policy in contract	Product
R10	Regulators do not accept Hydris PDFs as primary records	Medium	High	Original scan inside every PDF; e-signatures (FR-068); early SPCB conversations	Product
R11	Device refresh token in browser storage on the web glasses app	Low	Medium	Native app with Keystore is the pilot path	Security
R12	Audit page images stored in the database	Medium	Low	Move to object storage in Hardening	Platform

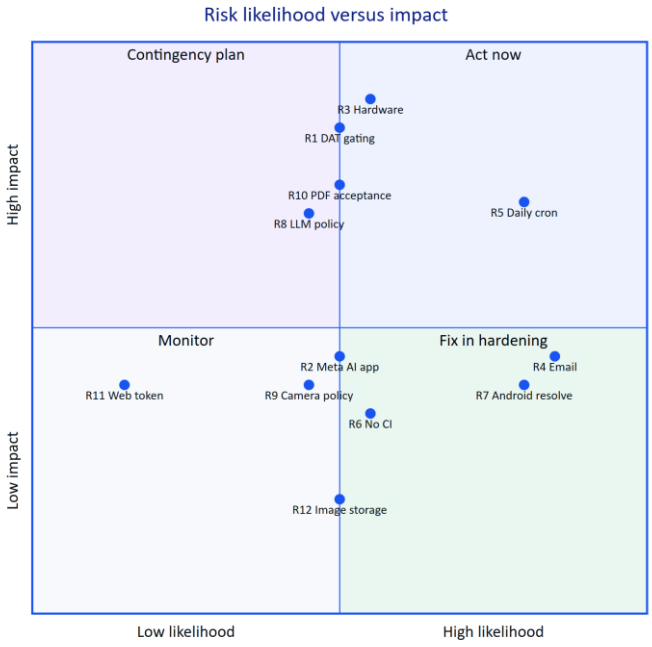


Figure 23. *Risk map by likelihood and impact (product team estimate)*

RISK

The largest schedule risk is R3 combined with R1: everything unique to Hydris on glasses (camera scanning, capture button, glasses microphone) is code-complete but unproven on hardware, and distribution beyond Developer Mode is in Meta's hands. The phone-first decision protects the pilot.

17 Open questions and decisions

Table 51. Open questions with owner and due point

ID	Question	Owner	Due
Q1	Is India among Meta's supported countries for full DAT capabilities and Developer Center access? Written confirmation needed	Product	Before hardware validation
Q2	When do DAT store submissions open, and what review does an unlisted enterprise app need?	Product	GA planning
Q3	Will pilot customers allow camera glasses in their plants, and in which zones?	Customer success	Pilot contract
Q4	Which regulators accept a Hydris PDF as the primary record rather than a scanned copy?	Product, pilot customer	Pilot exit
Q5	Is an iOS app needed for pilot or GA customers?	Product	GA planning
Q6	Must LLM inference run in India for Indian customers, and through which provider?	Security	Before enabling the model for a customer
Q7	Does acknowledgement on the glasses count as the assigned operator or the device in compliance reports?	Product, compliance	Pilot
Q8	What consent-limit data model does the compliance summary (FR-120) need?	Product	Pilot
Q9	Which email provider meets deliverability and residency needs?	Platform	Hardening
Q10	Should the handover summary (FR-122) use the model or deterministic templates?	Product	GA planning

Table 52. Decisions recorded in this PRD

Decision	Rationale
Hydris is read-only towards the plant	Keeps Hydris outside the control-system safety case
The pilot runs on the native Android app, not the WebView bridge	Keystore token storage, on-device OCR, direct DAT access
Every glasses feature works on the phone first	Decouples the pilot from hardware and Meta distribution risk
Rules engine and OCR-only modes are first-class	Pilots can run without the model until a tenant consents to processing
Approval needs a second, different person	Records stand up to an audit (P5)

18 Appendix

18.1 Glossary

Table 53. Terms used in this document

Term	Meaning
ETP, STP, WTP, CETP	Effluent, sewage, water and common effluent treatment plant
CPCB, SPCB	Central and State Pollution Control Boards in India
OCEMS	Online continuous effluent and emission monitoring system
Consent to operate	SPCB permission that sets a plant's discharge limits
Measurand	A quantity a sensor measures, such as pH or dissolved oxygen
Freshness	Live, delayed, stale or no data, from a sensor's expected interval
Deadband, on-delay	Hysteresis band and persistence time that stop alarms chattering
RLS	Row-level security in Postgres, used for tenant isolation
SSE	Server-sent events, the realtime channel to clients
DAT	Meta Wearables Device Access Toolkit
Meta AI app	Meta's companion app that pairs, registers and updates the glasses
UI profile	Versioned configuration of what a pair of glasses shows and allows
Four eyes	A second, different person approves what the first verified
Agreement	Whether independent OCR and vision readings of a value match
Demo mode	Zero-configuration deployment with a synthetic tenant
Built, Partial, Planned	On `main` and exercised; partly built or not validated; not yet in the repository

18.2 Related documents

Table 54. Related documents in docs/product/

Document	What it covers
`PRD.md`	Full requirements with every FR and NFR, acceptance criteria and priorities
`book/`	The product book: one compact PRD per product
`ARCHITECTURE.md`	Containers, flows, tenancy and token model
`DATA_MODEL.md`	Tables, enums, RLS and retention
`API_REFERENCE.md`	Routes, payloads, errors and limits
`MOBILE_APP_SPEC.md`	Hydris Field Android specification
`GLASSES_INTEGRATION.md`	Ray-Ban Meta and DAT capability matrix and flows
`DESIGN_SYSTEM.md`	Tokens, heads-up display and console patterns
`SECURITY_PRIVACY.md`	Controls, threat model and data handling
`DEPLOYMENT.md`	Topologies, environment and distribution
`QA_TEST_PLAN.md`	Test strategy, hardware matrix and journeys
`OPERATIONS_RUNBOOK.md`	Onboarding, incidents, backups and support

Document	What it covers
`ROADMAP.md`	Horizons, timeline and future products